

codex-windows / 019f2174-3a9f-7763-bd22-53158b567499

Metadata

```
- Source: `codex-windows`  
- Kind: `codex`  
- Source file: `C:\Users\avidu\.codex\archived_sessions\rollout-2026-07-02T11-41-33-019f2174-3a9f-7763-bd22-53158b567499.jsonl`  
- SHA-256: `c52f7e1fbf105a5f1154279895f61381e347f91620833cf6a31d70de4204b2cb`  
- Source modified: `2026-07-02T06:11:33+00:00`  
- Imported at: `2026-07-05T16:48:16+00:00`  
- cli_version: `0.142.5`  
- cwd: `C:\Users\avidu\Projects\badminton-highlight-indexer`  
- model_provider: `openai`  
- session_id: `019f2174-3a9f-7763-bd22-53158b567499`  
- source: `vscode`
```

Transcript

1. user (2026-07-02T06:11:33.152Z)

Confirm the just-landed Gate-A default flip (PR #146, merged to master as 570cf2b) causes no served-side regression on the real fusion path.

Context: PR #146 flipped the served default to `fusion\_hybrid` with `indexing.fusion.min\_crossings=2` (Gate-A held-shuttle guard, net_crossings>=2). The flip is validated OFFLINE (LOVO ablation, n=6 golden: ?F1 +0.074, CI [+0.042,+0.105], 6/6, p=0.031) and unit-tested, but the production `FusionSegmenter` served path (full WASB-via-WSL + Gemini handoff) has only ever been mock-tested on real video ? its byte-identity to `wasb\_hybrid` at `min\_crossings=0` was verified by #117's review but never run end-to-end with Gemini.

Task: on the GPU machine (torch+CUDA, WSL Ubuntu WASB, Gemini key in .env all present), run ONE or a few golden videos through the live `fusion\_hybrid` segmenter two ways ? `indexing.fusion.min\_crossings=0` (control = ungated, should reproduce wasb_hybrid) and `min\_crossings=2` (Gate-A default) ? and confirm: (a) the control run's served rallies match a `wasb\_hybrid` run on the same video (byte-identical seeding/handoff at mc=0); (b) the Gate-A run drops only low-crossing (held-shuttle/feed) windows, no real-rally loss vs the golden labels (or at least no F1 regression). Use the existing offline-rescore tooling where possible to avoid re-running WASB (wasb_runner has NO cache-hit early-return ? re-running overwrites the cached CSV; re-score the cached `output/wasb/\*\_wasb.csv` instead, see scratch/run_gx010125_rescore.py and backend/eval/distill_local.build_candidates). Keep artifacts isolated (per-video output dir). Record the result in docs/QUALITY_ITERATIONS.md (honest measured numbers only). If a regression appears, the rollback is a single flag: set `indexing.fusion.min\_crossings: 0` (Gate-A off) or `indexing.default\_segmenter: wasb\_hybrid` (full revert) in config.json.

This is the optional owner-gated follow-up flagged in PR #146 ? it does NOT change the merge decision; it adds served-side confidence.

2. assistant (2026-07-02T06:11:33.153Z)

I'll start by orienting myself: pulling the repo, reading the current-state/handoff, and checking what's actually available on this machine (the task assumes a GPU machine with torch+CUDA+WSL+Gemini).

3. assistant (2026-07-02T06:11:33.153Z)

```
[external_agent_tool_call: Bash]  
description: Fast-forward pull and show branch/HEAD  
command: git pull --ff-only 2>&1 | head -40 && echo "---BRANCH---" && git branch --show-current && echo "---HEAD---" && git log --oneline -3  
[/external_agent_tool_call]
```

4. assistant (2026-07-02T06:11:33.153Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md
[/external_agent_tool_call]
```

5. assistant (2026-07-02T06:11:33.153Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\session-handoff.md
[/external_agent_tool_call]
```

6. assistant (2026-07-02T06:11:33.153Z)

```
[external_agent_tool_result]
Already up to date.
---BRANCH---
master
---HEAD---
e75ec35 feat(eval): LOS0/LOG0 ablation runner ? per-signal marginal ?F1 (Phase 0) (#147)
570cf2b feat(fusion): flip Gate-A (held-shuttle guard) to the served default ? config-reversible
(#146)
7c6498f docs(quality): personalization Phase-0 checkpoint ? eval/safety rails + locked
structural-guard decision (#145)
[/external_agent_tool_result]
```

7. assistant (2026-07-02T06:11:33.153Z)

```
[external_agent_tool_result]
1      ---
2      name: session-handoff
3      description: "Living \"you are here\" handoff ? current state, next steps, open
decisions, ramp-up kit. AUTO-REFRESHED at meaningful checkpoints (PR merged, key decision, phase
shift, session wrap). \"update the handoff\" forces a full rewrite."
4      metadata:
5          node_type: memory
6          type: project
7          originSessionId: 45ea3a91-654d-4972-9cd3-65f35db54caf
8      ---
9
10     # Session handoff ? sports highlight indexer (repo still named badminton-highlight-
indexer; RENAME PENDING) + siblings: sports-data-collector, rally-annotator, sports-obsreport
11
12     **Last updated:** 2026-06-14 (session wrap; master at #135; self-merged on green CI
throughout). **?? IMMEDIATE PICKUP =
13     EXECUTION-POLICY P3 (then P4)** ? a per-job, self-scheduling resilience framework
(owner's design, validated). **DESIGN
14     ([#132](https://github.com/avidullu/badminton-highlight-indexer/pull/132)
`docs/EXECUTION_POLICY.md`) + P1 the actual
15     Gemini-hang fix ([#133](https://github.com/avidullu/badminton-highlight-
indexer/pull/133)
16     `backend/infrastructure/execution_policy.py` ? `run_bounded`/`poll_until`, 100% module
coverage) + P2 the self-scheduling
17     DB substrate ([#135](https://github.com/avidullu/badminton-highlight-indexer/pull/135) ?
jobs migration v3 `policy`/`next_poll_at`,
18     `claim_due_job`/`set_job_waiting`) are MERGED.** **NOT DONE (next session, own distinct
PRs, HIGH coverage):** **P3** =
19     wire the `main.py` executor to a `claim_due_job` self-scheduling loop honoring
WAIT_AND_RESUME (release+reschedule) +
20     **resumable AI handoff** (persist per-window confirmation; resume only unconfirmed ? the
GX010125 reel becomes exhaustive
21     across resumes, no wasted Gemini spend); **P4** = apply the policy to the WSL/WASB GPU
pass + ffmpeg. **Read [[current-state]]
22     FIRST** (the top checkpoint has the full P1/P2 detail + the rationale for stopping at
```

```

P2).
23
24     **Earlier this same session (rally-quality / paper track, [[paper-coauthor-aim]]):**
analyzers/reconcilers **framework**
25     doc + a verified **multi-agent literature review** (§13 ? skeleton is standard
TAL/TAS+late-fusion, *cite don't claim*;
26     only the report-first reconciler is a novel combination) ? [#119]; **eval measurement-
honesty helpers C1?C4** (`backend/eval/
27     eval_stats.py`+`score_calibration.py`+`segmentation_metrics.py`) ? [#124] + wired into
`experiment.compare` ? [#129];
28     **FIRST golden A/B experiment** (n=6, offline) ? archived [#128]: **Gate-A (net-
crossings?2) is a measured win (+19% F1,
29     6/6, p=0.031), PROMOTED** [#131]; the window-level **learned variant FIXED** (train-fold
threshold tuning + calibration ?
30     +38% F1, un-zeroed) ? [#130]; **`docs/PER_VIDEO_OUTPUT_LAYOUT.md`** plan [#131];
**GX010125_1080p reel re-generated** into
31     isolated `output/GX010125_run/` (reused cached 1080p trajectory). Prior tracks still
valid: rally-quality fusion P1?P4
32     (#114?127), UI-alpha, productization, first 4K-GPU E2E. This is the ~1-page orientation.
? **THIS IS THE OWNER'S REAL GPU
33     MACHINE** (torch+CUDA/cv2/WSL/Gemini live).
34
35     ## You are here (2026-06-13) ? ? HARDENED + M0 SCAFFOLDED + MOAT QUANTIFIED + QUALITY-
SWEPT + UI-ALPHA LANDING + 4K-E2E PROVEN
36     -1. **Recent landings since the sweep (mostly OTHER slates ? facts from git/gh, not this
session's work):**
37     - **UI-alpha:** PR-1 async jobs (#87) + **PR-2 chunked upload + highlight download
(#99, live-E2E-verified)** +
38     docs sync (#100). Next UI slice = **PR-3 identity** (`owner_email` scoping,
JWT/header w/ `DEV_USER_EMAIL`
39     fallback, CORS narrow, rate limit). ? PR-3/deploy: `security.allowed_video_root`
MUST contain `upload_dir`.
40     - **Branding/stitcher:** #101 configurable watermark · #102 config paddings ? live
VideoCompiler · #103 bundled
41     Apache-2.0 font + fallback log · #107 `/api/compile` enforces
`stitching.min_shot_count` · #108 escape Windows
42     drive-letter colon in ffmpeg filtergraph (fixed #103's watermark silently failing
on Windows...
[/external_agent_tool_result]

```

8. assistant (2026-07-02T06:11:33.153Z)

```

[external_agent_tool_result]
<system-reminder>[Truncated: PARTIAL view ? showing lines 1-451 of 1851 total (87024 tokens, cap
25000). Call Read with offset=452 limit=451 for the next page, or Grep to find a specific
section. Do NOT answer from this page alone if the answer may be further in the file.]</system-
reminder>

```

```

1     ---
2     name: current-state
3     description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
only on request). What just landed, what's next, open decisions."
4     metadata:
5         node_type: memory
6         type: project
7         originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8     ---
9
10    # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12    **Checkpoint:** 2026-06-14 (PERSONALIZATION Phase-0 LANDED + quality checkpoint).
**Owner adopted the HYBRID** ? personalization as a
13    flag-gated FEATURE on a generalization-first baseline + a **contribution flywheel** ("a
tool that gets better the more you help it";
14    free-tier videos pool into the owner-trained baseline). Pivot analysis

```

```
(`scratch/PERSONALIZATION_PIVOT_ANALYSIS.md`) recommended
15     hybrid-not-personalization-first; the owner's contribution-flywheel closes the flywheel-
loss objection. **MERGED this session:**
16     telemetry wire-in **#140**; per-user promotion gate **#141** (min-N floor + structural-
guard exemption); held-out-USER learning
17     curve **#142** (" $K$  videos to superior quality" = falsifiable); per-user no-regression
guard **#143**; **`docs/PERSONALIZATION_PLAN.md`
18     #144** (the checked-in project doc + locked decisions). Gemini-training-source concern =
**CLEAN** (eval-only, gitignored, M0.3 owner-gated).
19     **LOCKED owner decisions:** identity = north-star-design/current-impl (nullable user_id,
device UUID); inline classifier_json behind a
20     cloud-load interface; **min_n` is a PROMOTION-confidence floor, NOT a service gate** (a
1-video user is always fully segmented by the
21     baseline); free-tier video pooling (consent + minors-exclusion); **Gate-A-only
structural + `min_n=5`**. ?? **IN FLIGHT (3 bg agents):**
22     quality-iterations doc PR; **Gate-A/B golden-set numbers + flip-implications**
(`scratch/GATE_AB_NUMBERS_AND_IMPLICATIONS.md` ? leave-one-
23     signal-out ablation; owner: NUMBERS INFORM, FOUNDATIONS DECIDE); **ablation-layer
framework plan** (`scratch/ABLATION_FRAMEWORK_PLAN.md` ?
24     leave-one-signal-out + signal-relevance ledger over time, for the paper). **GPU + Gemini
API now AUTHORIZED for evals** (cooler boost on).
25     **Gate-A default flip = NOT done** ? gather numbers + implications first, then owner
decides (data-influenced, foundations-led). NEXT
26     personalization = Phase 1 (serving bridge: wire `LogisticRegression.load` into
`select_for_handoff`) ? owner-gated, see [[weak-signals-retained]] +
`docs/PERSONALIZATION_PLAN.md`.
27
28     **Checkpoint:** 2026-06-14 (FUSION P3/P4 MEASURED + extraction COMPLETE + telemetry #134
merged + Cooler Boost). **THE NUMBERS
29     (honest LOVO n=6, IoU $\geq$ 0.5, C1 bootstrap-CI + sign-test):** shuttle-only F1= $\approx$ 0.307**;
**+person (P3) ?F1= $\approx$ 0.021** (?6.7%, CI
30     [ $\approx$ 0.165, $\approx$ 0.162], sign 2W/4L p=0.688, **CI-clears-0: NO**) ? NO lift, stays default-OFF
(no court polygons ? `players_in_court`
31     counts spectators; per-video wash: testlarge +0.40 / gbaaddy  $\approx$ 0.22). **+person+audio (P4
incremental) ?F1= $\approx$ 0.079** (+27.7%, F1
32     0.286? $\approx$ 0.366**, CI [ $\approx$ 0.054, $\approx$ 0.203], sign 4W/2L p=0.688, **CI-clears-0: NO**) ?
PROMISING (wins 4/6, largetest/mahadevpura
33     +0.27; endpoint 0.366 > shuttle-only 0.307) but n=6 can't confirm. **VERDICT: neither
promotable (both CIs span 0); audio is
34     the lead ? need more matches + measure audio on shuttle-only directly (drop the person
handicap).** Recorded in
35     `eval_baselines/experiment_leaderboard.json`; honest negative/uncertain results KEPT
([[paper-coauthor-aim]]). **LANDED THIS
36     SESSION:** result logged to `docs/QUALITY_ITERATIONS.md` (#137);
`experiment_leaderboard.json` tracked (#139); **weak-signals
37     audit ?** (person/audio/court-bounds all present, de...
[/external_agent_tool_result]
```

9. assistant (2026-07-02T06:11:33.153Z)

This is the owner's real GPU machine, and the task is well-scoped. Let me orient on the tooling and environment before doing anything. Let me read the key files and check what's cached.

10. assistant (2026-07-02T06:11:33.153Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\run_gx010125_rescore.py
[/external_agent_tool_call]
```

11. assistant (2026-07-02T06:11:33.153Z)

```
[external_agent_tool_result]
1     """Run the framework on GX010125_1080p by REUSING its cached WASB trajectory ? offline,
2     no GPU, no 4K, isolated outputs (output/GX010125_run/), zero pollution of past work.
```

```

3
4     This is the "persist detection once, re-score forever" path on a REAL, unlabeled match:
5     parse the cached 1080p trajectory -> candidate rally windows -> CONTROL vs Gate-A
6     (net-crossings >= 2, the measured precision win) side-by-side (no golden labels exist
for
7     this footage, so we report the divergence, not F1).
8     ""
9     import json
10    import os
11
12    from backend.eval.distill_local import build_candidates, FEATURE_NAMES
13    from backend.eval.experiment import CONTROL, Variant, VideoCandidates,
compare_unlabeled, predict
14
15    TRAJ = "output/wasb/GX010125_1080p__89605e80ed01_wasb.csv"
16    FPS, FRAME_W = 59.94, 1920.0
17    OUT_DIR = "output/GX010125_run"
18
19
20    def _dur(ivs):
21        return sum(e - s for s, e in ivs)
22
23
24    def _write_csv(path, ivs):
25        with open(path, "w", encoding="utf-8") as f:
26            f.write("start_sec,end_sec,duration_sec\n")
27            for s, e in ivs:
28                f.write(f"{s:.3f},{e:.3f},{e - s:.3f}\n")
29
30
31    def main():
32        os.makedirs(OUT_DIR, exist_ok=True)
33        intervals, X = build_candidates(TRAJ, FPS, FRAME_W)
34        feats = {FEATURE_NAMES[k]: [r[k] for r in X] for k in range(len(FEATURE_NAMES))}
35        vc = VideoCandidates(name="GX010125_1080p", intervals=intervals, features=feats,
gts=None)
36
37        gateA = Variant(name="gateA", min_crossings=2)
38        ctrl_preds = predict(CONTROL, vc)
39        gate_preds = predict(gateA, vc)
40        sxs = compare_unlabeled(vc, CONTROL, gateA) # agreed / a_only (dropped by Gate-A)
/ b_only
41
42        print("=" * 70)
43        print(f"GX010125_1080p ? offline re-score of the CACHED trajectory (no GPU)")
44        print("=" * 70)
45        print(f" candidate windows from trajectory : {len(intervals)}")
46        print(f" CONTROL rallies (merge all)          : {len(ctrl_preds)} ({_dur(ctrl_preds)
/ 60:.1f} min)")
47        print(f" Gate-A rallies (net_crossings>=2) : {len(gate_preds)} ({_dur(gate_preds)
/ 60:.1f} min)")
48        print(f" Gate-A dropped (a_only)              : {len(sxs['a_only'])} windows "
49              f"({_dur(sxs['a_only']) / 60:.1f} min of likely held-shuttle/feed FPs)")
50        print(f" both agree (kept)                      : {sxs['num_agreed']}")
51
52        _write_csv(os.path.join(OUT_DIR, "GX010125_control_rallies.csv"), ctrl_preds)
53        _write_csv(os.path.join(OUT_DIR, "GX010125_gateA_rallies.csv"), gate_preds)
54        summary = {
55            "video": "GX010125_1080p", "source": "cached WASB trajectory (offline re-score,
no GPU)",
56            "fps": FPS, "num_candidates": len(intervals),
57            "control": {"num_rallies": len(ctrl_preds), "total_min": round(_dur(ctrl_preds)
/ 60, 2)},
58            "gateA": {"num_rallies": len(gate_preds), "total_min": round(_dur(gate_preds) /
60, 2),

```

```

59         "dropped_windows": len(sxs["a_only"])),
60     }
61     json.dump(summary, open(os.path.join(OUT_DIR, "GX010125_rescore_summary.json"),
62 "w"), indent=2)
63     print(f"\n wrote isolated outputs -> {OUT_DIR}/ (control + gateA rally CSVs +
summary)")
64     print(" NOTE: no golden labels for this footage -> divergence reported, not F1. A
full")
65     print(" Gemini-confirmed highlight reel is a separate ~tens-of-min GPU+API run.")
66
67     if __name__ == "__main__":
68         main()
69
[/external_agent_tool_result]

```

12. assistant (2026-07-02T06:11:33.153Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\distill_local.py
[/external_agent_tool_call]

```

13. assistant (2026-07-02T06:11:33.153Z)

```

[external_agent_tool_result]
1     """Distill Gemini (teacher) into a LEARNED local rally classifier (student).
2
3     Threshold-tuning the local knobs hit a low ceiling (see calibrate_local). This goes
4     further: learn "is this WASB candidate a real rally?" from Gemini silver labels,
5     using a few resolution/fps-INVARIANT local features, so the model transfers across
6     videos and runs 100% local at inference (no Gemini, privacy lever intact).
7
8     Flow (all local at runtime):
9         WASB trajectory -> recall-oriented candidate windows (propose many; classifier
10 filters) -> per-candidate features [duration, peak_velocity, mean_velocity,
11 active_ratio, net_crossings] -> LogisticRegression(prob >= threshold) -> merge
12 -> rally windows.
13
14 Training labels come from Gemini full-context labels via the SAME IoU matching the
15 evaluator uses (training.label_candidates). Honest leave-one-video-out: train on the
16 other videos' candidates, predict the held-out one, score vs its Gemini labels, and
17 compare against the threshold baseline. Reuses classifier.py, training.label_candidates,
18 rally_gate, metrics, calibrate_wasb.load_golden_gts, gemini_refine.merge_overlaps.
19
20 Run:
21     python -m backend.eval.distill_local --manifest videos.json --model-out
output/local_rally_model.json
22     """
23     from __future__ import annotations
24
25     import json
26     import argparse
27     from typing import Dict, List, Optional, Tuple
28
29
30     from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
31     from backend.pipeline.detectors import rally_gate
32     from backend.eval.calibrate_wasb import load_golden_gts
33     from backend.eval.training import label_candidates
34     from backend.eval.classifier import LogisticRegression
35     from backend.eval.metrics import score
36     from backend.eval.gemini_refine import merge_overlaps
37
38     Interval = Tuple[float, float]
39

```

```

40     FEATURE_NAMES = ["duration", "peak_velocity", "mean_velocity", "active_ratio",
"net_crossings"]
41     # Recall-oriented proposal: deliberately permissive so real rallies are among the
42     # candidates (the classifier removes the junk). (velocity_thresh, merge_gap, min_dur)
43     PROPOSAL = (0.005, 1.0, 0.5)
44     BASELINE_LOCAL = (0.02, 2.0, 2.0)    # today's production-ish windowing, no learning
45
46
47     def build_candidates(trajecory_csv: str, fps: float, frame_width: float,
48                         proposal: Tuple[float, float, float] = PROPOSAL) ->
Tuple[List[Interval], List[List[float]]]:
49         """WASB trajectory -> (candidate intervals, fps/resolution-invariant feature
rows)."""
50         points = TrackNetRunner.parse_trajectory_csv(trajecory_csv)
51         vt, mg, mwd = proposal
52         wins = TrackNetRunner.trajectory_to_action_windows(
53             points, fps=fps, frame_width=frame_width,
54             velocity_thresh=vt, merge_gap=mg, min_window_duration=mwd)
55         intervals = [(w["start"], w["end"]) for w in wins]
56         gated = rally_gate.gate_windows(points, intervals, fps, min_crossings=0) # for net-
crossings
57         X: List[List[float]] = []
58         for w, g in zip(wins, gated):
59             dur = max(1e-6, w["end"] - w["start"])
60             win_frames = max(1.0, dur * fps)
61             X.append([
62                 dur,                                     # rally length (sec) ?
63                 float(w["peak_velocity"]),               # already normalized by
64                 float(w["mean_velocity"]),
65                 float(w["active_frame_count"]) / win_frames, # active-shuttle ratio ?
66                 float(g.crossings),                     # net crossings (count) ?
67                 ])
68         return intervals, X
69
70
71     def baseline_windows(trajecory_csv: str, fps: float, frame_width: float) ->
List[Interval]:
72         points = TrackNetRunner.parse_trajectory_csv(trajecory_csv)
73         vt, mg, mwd = BASELINE_LOCAL
74         wins = TrackNetRunner.trajectory_to_action_windows(
75             points, fps=fps, frame_width=frame_width,
76             ...
[/external_agent_tool_result]

```

14. assistant (2026-07-02T06:11:33.153Z)

```

[external_agent_tool_call: Bash]
description: List cached WASB CSVs and output dirs
command: echo "=== output/wasb CSVs ===" && ls -la output/wasb/ 2>/dev/null | head -40; echo
"=== output dirs ===" && ls output/ 2>/dev/null | head -50
[/external_agent_tool_call]

```

15. assistant (2026-07-02T06:11:33.154Z)

```

[external_agent_tool_result]
=== output/wasb CSVs ===
total 1804
drwxr-xr-x 1 avidu 197609      0 Jun 13 14:43 .
drwxr-xr-x 1 avidu 197609      0 Jun 14 11:51 ..
-rw-r--r-- 1 avidu 197609 1635249 Jun 13 14:43 GX010125_1080p__89605e80ed01_wasb.csv
drwxr-xr-x 1 avidu 197609      0 Jun 13 14:42 GX010125_1080p__89605e80ed01_wasb_wasbcache

```

```

-rw-r--r-- 1 avidu 197609 172447 Jun 11 18:28 mahadevpura_smoke_180s__05e0e577029d_wasb.csv
drwxr-xr-x 1 avidu 197609 0 Jun 11 18:28
mahadevpura_smoke_180s__05e0e577029d_wasb_wasbcache
=== output dirs ===
GX010125_1080p.mp4
GX010125_highlights.mp4
GX010125_highlights_min5.mp4
GX010125_run
KushagraAviHighlight-Trailor.mp4
MahadevpuraSingles_highlights.mp4
ab_experiment_n6.json
adarsh_avi_full_vs_human.csv
adarsh_avi_singles_golden_features.json
adarsh_avi_vs_human.csv
adarsh_baseline_gated.csv
adarsh_baseline_segments.csv
adarsh_gemini_fullvideo.csv
adarsh_gemini_refined.csv
adarsh_gemini_smoke.csv
adarsh_min5s_segments.csv
adarsh_tuned_segments.csv
badminton_indexer.db
e2e_reel.mp4
e2e_server_log.txt
fusion_golden.log
gbaaddy_golden_features.json
gbaaddy_vs_human.csv
gx010125_min5_stitch.log
gx010125_run.log
gx010125_transcode.log
human_lovo_manifest.json
kushagra_1280.mp4
kushagra_1280.report.json
kushagra_1280.report.md
kushagra_1280.summary.md
kushagra_singles_golden_features.json
kushagra_vs_human.csv
largetest_doubles_golden_features.json
largetestvideo_vs_human.csv
local_calib_manifest.json
local_rally_model.json
mahadevpura_gemini_refined.csv
mahadevpura_singles_golden_features.json
mahadevpura_smoke_180s.mp4
mahadevpura_smoke_180s.report.json
mahadevpura_smoke_180s.report.md
mahadevpura_smoke_180s.summary.md
mahadevpura_vs_human.csv
phase3_probe.json
phase3_yolo.db
ramp_probe.mp4
smoke4_log.txt
sports_indexer.db
test_indexer.db
[/external_agent_tool_result]

```

16. assistant (2026-07-02T06:11:33.154Z)

I have cached WASB CSVs for GX010125 (unlabeled) and a mahadevpura smoke clip. The golden corpus (with labels) is referenced by the manifest. Let me read the manifest, the FusionSegmenter served path, the experiment harness, and the existing n6 LOVO repro ? plus check the environment.

17. assistant (2026-07-02T06:11:33.154Z)

```
[external_agent_tool_call: Bash]
description: Read golden manifest and indexing config
command: echo "=== human_lovo_manifest.json ===" && cat output/human_lovo_manifest.json
2>/dev/null; echo; echo "=== config.json (indexing section) ===" && python -c "import
json;c=json.load(open('config.json'));print(json.dumps(c.get('indexing',{}),indent=2))" 2>&1 |
head -60
[/external_agent_tool_call]
```

18. assistant (2026-07-02T06:11:33.154Z)

```
[external_agent_tool_result]
=== human_lovo_manifest.json ===
[
  {
    "name": "adarsh_avi_singles",
    "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/adarsh_avi_full_traj.csv",
    "fps": 59.94, "frame_width": 1920,
    "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden Labelled/Adarsh and
Avi.rallies.csv"
  },
  {
    "name": "kushagra_singles",
    "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/kushagra_traj.csv",
    "fps": 59.94, "frame_width": 1920,
    "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden
Labelled/KushagraYashAviFirstVideo.rallies.csv"
  },
  {
    "name": "largetest_doubles",
    "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/largetestvideo_traj.csv",
    "fps": 29.97, "frame_width": 1920,
    "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden
Labelled/LargeTestVideo.rallies.csv"
  },
  {
    "name": "gbaaddy",
    "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/gbaaddy_traj.csv",
    "fps": 59.94, "frame_width": 1920,
    "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden
Labelled/gBaaddy20May2026.rallies.csv"
  },
  {
    "name": "testlarge_short",
    "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/testlarge_traj.csv",
    "fps": 29.97, "frame_width": 1920,
    "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden
Labelled/TestLargeVideo.rallies.csv"
  },
  {
    "name": "mahadevpura_singles",
    "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/mahadevpura_traj.csv",
    "fps": 29.97, "frame_width": 1920,
    "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden
Labelled/MahadevpuraSingles.rallies.csv"
  }
]

=== config.json (indexing section) ===
{
```

```

"default_segmenter": "fusion_hybrid",
"use_gpu": true,
"motion_threshold": 0.08,
"min_segment_duration": 3.0,
"dead_time_merge_gap": 2.0,
"chunk_duration": 90,
"chunk_overlap": 10,
"detector_model": "fasterrcnn_resnet50_fpn",
"detector_score_threshold": 0.5,
"yolo_velocity_threshold": 0.15,
"yolo_frame_skip": 3,
"yolo_tracker": "bytetrack.yaml",
"yolo_prefetch_workers": 2,
"min_action_window_duration": 2.0,
"log_yolo_candidates": true,
"store_yolo_candidates": true,
"ai_handoff_padding": 2.0,
"ai_max_workers": 1,
"tracknet": {
    "wsl_distro": "Ubuntu",
    "repo_dir": "~/models/TrackNetV4",
    "conda_sh": "~/miniconda3/etc/profile.d/conda.sh",
    "conda_env": "TrackNetV4",
    "weights_path": "",
    "queue_length": 5,
    "stage_in_wsl": true,
    "wsl_stage_dir": "~/clips",
    "timeout_sec": 1800,
    "velocity_threshold": 0.02
},
"wasb": {
    "wsl_distro": "Ubuntu",
    "repo_dir": "~/models/WASB-SBDT",
    "conda_sh": "~/miniconda3/etc/profile.d/conda.sh",
    "conda_env": "wasb",
    "weights_path": "~/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar",
    "sport": "badminton",
    "wsl_stage_dir": "~/clips",
    "timeout_sec": 1800,
    "velocity_threshold": 0.02
},
"fusion": {
    "substrate": "wasb",
    "min_crossings": 2,
    "min_players": 0,
    "cue_flags": {}
}
}
[/external_agent_tool_result]

```

19. assistant (2026-07-02T06:11:33.154Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\fusion_hybrid.py
[/external_agent_tool_call]

```

20. assistant (2026-07-02T06:11:33.154Z)

```

[external_agent_tool_result]
1      """Multi-signal fusion segmenter (MULTI_SIGNAL_FUSION_PLAN P2).
2
3      Shuttle stays PRIMARY: this reuses an existing trajectory substrate (WASB by default,
4      TrackNet optional) to seed candidate windows exactly as `wasb_hybrid`/`tracknet_hybrid`
5      do ? it subclasses the same `TrajectoryHybridSegmenter` engine. It then ENRICHES each

```

```

6      shuttle-seeded window via two overridable seams the engine exposes:
7
8      - ``_enrich_candidate_stats`` ? always adds the **net-crossing count** (Gate A signal,
9        GPU-free, computed from the trajectory we already have); and, only when the person cue
10       is explicitly enabled (``indexing.fusion.cue_flags.person``), the person-cue features
11       (``fusion_features``), persisting them into ``candidate_segments.stats`` for the learned
12       fusion (P3). The person path lazily builds ONE ``PersonDetector`` and only then loads
the
13       torchvision/supervision model ? so the default (person-OFF) path never loads a
detector
14       model nor touches the GPU. (torch/cv2 themselves are imported by the registry
regardless,
15       via the pre-existing yolo_hybrid ? person_detector chain; the lazy part is the model.)
16       - ``_select_for_handoff`` ? optional served Gate A/B: when
``indexing.fusion.min_crossings``
17       (and, with the person cue, ``min_players``) are raised above 0, sub-threshold windows
are
18       dropped from the AI handoff (the held-shuttle / one-sided-feed false-positive fix).
19       **Persisted candidates remain the full superset** regardless, so the offline
experiment
20       harness can re-score any variant.
21
22       Defaults reproduce the substrate exactly: net_crossings/person features are persisted
but
23       nothing is gated (thresholds 0, person OFF), so ``FusionSegmenter`` with default config
seeds
24       and hands off identically to ``wasb_hybrid``. Registered as ``fusion_hybrid``, NOT the
default
25       segmenter ? it is opt-in. Promotion to default happens only on measured LOVO lift
through the
26       experiment harness (EXPERIMENT_HARNESS.md), never silently.
27       """
28       from typing import Any, Dict, List, Optional, Tuple
29
30       from backend.pipeline.segmenters.base import segmenter_registry
31       from backend.pipeline.segmenters.trajectory_hybrid import TrajectoryHybridSegmenter
32       from backend.pipeline.detectors.base import DetectorRunner
33       from backend.pipeline.detectors.wasb_runner import WasbRunner, WasbConfig
34       from backend.pipeline.detectors.tracknet_runner import TrackNetRunner, TrackNetConfig
35       from backend.pipeline.detectors import rally_gate
36       from backend.pipeline.detectors import fusion_features as ff
37
38
39       class FusionSegmenter(TrajectoryHybridSegmenter):
40           """Shuttle trajectory (primary) + net-crossing Gate A + optional person cue (Gate
B)."""
41
42           family = "fusion"
43           display_label = "Fusion"
44
45           def __init__(self, db: Any, config: Any):
46               super().__init__(db, config)
47               # Built lazily only if the person cue is enabled (no detector model loaded
otherwise).
48               self._person: Optional[Any] = None
49               # Cache the play-axis/net estimate per trajectory so rally_gate doesn't re-
estimate it
50               # over the whole trajectory once per candidate window (it depends only on
``points``).
51               self._axis_cache_key: Optional[int] = None
52               self._axis_cache: Optional[tuple] = None
53
54           @property
55           def name(self) -> str:
56               return "fusion_hybrid"

```

```

57
58     @property
59     def description(self) -> str:
60         return ("Multi-signal fusion: shuttle trajectory (WASB/TrackNet) seeds candidate
rallies, "
61                 "net-crossing Gate A + optional person-occupancy Gate B adjudicate them,
and the "
62                 "AI provider sets semantic boundaries.")
63
64     def _build_runner(self, idx_cfg: Dict[str, Any]) -> Tuple[DetectorRunner, Dict[str,
Any]]:
65         substrate = idx_cfg.get("fusion", {}).get("substrate", "wasb")
66         if substrate == "tracknet":
67             cfg = TrackNetConfig.from_indexing...
[/external_agent_tool_result]

```

21. assistant (2026-07-02T06:11:33.154Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\ab_n6_lovo.py
[/external_agent_tool_call]

```

22. assistant (2026-07-02T06:11:33.154Z)

```

[external_agent_tool_result]
1     """First full A/B experiment on the n=6 golden corpus ? runs entirely offline
2     (persisted WASB trajectories, no GPU). One consistent candidate set per video
3     (recall-oriented proposal windowing, the persisted pipeline's own), re-scored three
4     ways, then graded with the C1 honest aggregation + C4 segmentation metrics + a C3
5     out-of-fold learned mAP. Prints a report and writes output/ab_experiment_n6.json.
6
7     Honest: prints whatever the real golden data produces. No claims before measurement.
8     """
9     import json
10
11     from backend.eval.distill_local import FEATURE_NAMES, build_candidates
12     from backend.eval.calibrate_wasb import load_golden_gts
13     from backend.eval.training import label_candidates
14     from backend.eval.classifier import LogisticRegression
15     from backend.eval.experiment import CONTROL, Variant, VideoCandidates, evaluate_variant
16     from backend.eval.metrics import score
17     from backend.eval import segmentation_metrics as sm
18     from backend.eval import eval_stats as es
19
20     IOU = 0.5
21     MANIFEST = "output/human_lovo_manifest.json"
22     OUT = "output/ab_experiment_n6.json"
23
24
25     def load_corpus():
26         specs = json.load(open(MANIFEST))
27         videos, pooled = [], {"X": [], "y": [], "group": [], "interval": []}
28         for s in specs:
29             fps, fw = float(s["fps"]), float(s["frame_width"])
30             intervals, X = build_candidates(s["trajectory"], fps, fw)
31             gts = load_golden_gts(s["labels"])
32             y = label_candidates(intervals, gts, IOU)
33             feats = {FEATURE_NAMES[k]: [row[k] for row in X] for k in
range(len(FEATURE_NAMES))}
34             videos.append(VideoCandidates(name=s["name"], intervals=intervals,
features=feats, gts=gts))
35             for iv, row, yi in zip(intervals, X, y):
36                 pooled["X"].append(row); pooled["y"].append(yi)
37                 pooled["group"].append(s["name"]); pooled["interval"].append(iv)
38         return videos, pooled

```

```

39
40
41     def c4_aggregate(ev, videos):
42         """Mean F1@k + mean segment-count ratio across videos for one variant's
43         predictions."""
44         by_name = {v.name: v for v in videos}
45         f1k = {0.1: [], 0.25: [], 0.5: []}
46         ratios = []
47         for name, preds in ev["predictions"].items():
48             gts = by_name[name].gts
49             got = sm.f1_at_overlaps(preds, gts)
50             for k in f1k:
51                 f1k[k].append(got[k])
52             ratios.append(sm.segment_count_ratio(preds, gts))
53         mean = lambda xs: sum(xs) / len(xs) if xs else 0.0
54         return {"f1@{int(k*100)}": round(mean(v), 3) for k, v in f1k.items()} | \
55             {"seg_count_ratio": round(mean(ratios), 2)}
56
57     def oof_learned_map(pooled, videos):
58         """C3 + C4: out-of-fold learned candidate proba ? mAP@tIoU per video, averaged."""
59         def fit_predict(train_idx, test_idx):
60             m = LogisticRegression().fit([pooled["X"][i] for i in train_idx],
61                                         [pooled["y"][i] for i in train_idx], FEATURE_NAMES)
62             return list(m.predict_proba([pooled["X"][i] for i in test_idx]))
63         oof = es.out_of_fold_predictions(pooled["group"], fit_predict)
64         by_name = {v.name: v for v in videos}
65         maps = []
66         for name in by_name:
67             scored = [(pooled["interval"][i], float(oof[i]))
68                       for i in range(len(oof)) if pooled["group"][i] == name and oof[i] is
69 not None]
70             maps.append(sm.mean_average_precision(scored, by_name[name].gts))
71         return sum(maps) / len(maps) if maps else 0.0
72
73     def c1_block(a_f1, b_f1):
74         d = es.paired_delta_ci(a_f1, b_f1, seed=0)
75         return {
76             "mean_a": round(sum(a_f1) / len(a_f1), 3),
77             "mean_b": round(sum(b_f1) / len(b_f1), 3),
78             "mean_delta": round(d["mean_delta"], 3),
79             "ci": [round(d["ci_low"], 3), round(d["ci_high"], 3)],
80             "ci_excludes_zero": d["ci_excludes_zero"],
81             "sign": {k: int(v) if k != "p_value" else round(v, 3) for k, v in
82 d["sign_test"].items()}},
83     8...
84 [/external_agent_tool_result]

```

23. assistant (2026-07-02T06:11:33.154Z)

Now I understand the architecture. The key realization: the offline harness (`build\_candidates` + `experiment.predict`) reimplements gating but does NOT exercise the real `FusionSegmenter.\_enrich\_candidate\_stats`/_select_for_handoff` served code. To close the served-side gap properly, I should drive the **actual** `FusionSegmenter` methods over the cached trajectory. Let me read the base engine to understand the production windowing path and whether I can drive it offline.

24. assistant (2026-07-02T06:11:33.154Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\trajectory_hybrid.py
[/external_agent_tool_call]

```

25. assistant (2026-07-02T06:11:33.154Z)

```
[external_agent_tool_result]
1      """Shared engine for trajectory-substrate hybrid segmenters.
2
3      tracknet_hybrid and wasb_hybrid were copy-paste twins ? CODE_MAP carried a
4      standing footgun: "any P3/P4 fix must be applied to BOTH". This base class is
5      now the single home for the whole pipeline shape:
6
7          healthcheck gate -> P2 trajectory -> candidate windows (+ persistence)
8          -> P3 AI-provider handoff over padded clips -> P4 DB population/linking
9
10     A concrete subclass supplies only its identity and runner wiring:
11     - ``family`` ? config section under ``indexing.*``, output subdir, and
12       the ``<family>-hybrid-<model>`` detector tag in metadata.
13     - ``display_label`` ? human label for log lines ("WASB", "TrackNet").
14     - ``name`` / ``description`` properties (the registry contract).
15     - ``_build_runner(idx_cfg)`` ? returns (DetectorRunner, detector-specific
16       detection_params extras such as weights_path).
17
18     This base is deliberately NOT registered; only concrete subclasses register.
19     """
20     import os
21     import time
22     import logging
23     import concurrent.futures
24     from typing import Any, Dict, List, Optional, Tuple
25
26     import cv2
27
28     from backend.pipeline.segmenters.base import BasePlaySegmenter
29     from backend.utils.video import get_video_duration, extract_video_chunk
30     from backend.sports import sport_registry
31     from backend.providers import provider_registry
32     from backend.pipeline.detectors.base import DetectorRunner
33
34     logger = logging.getLogger(__name__)
35
36
37     def _fmt_ts(seconds: float) -> str:
38         seconds = max(0, int(seconds))
39         h, rem = divmod(seconds, 3600)
40         m, s = divmod(rem, 60)
41         return f"{h:02d}:{m:02d}:{s:02d}"
42
43
44     class TrajectoryHybridSegmenter(BasePlaySegmenter):
45         """Trajectory-detector hybrid: precise candidate rallies + AI semantic
46         boundaries."""
47
48         #: config section under `indexing` (velocity_threshold lives there), the
49         #: output subdir for trajectories, and the metadata detector-tag prefix.
50         family: str = ""
51         #: human-readable label used in log lines.
52         display_label: str = ""
53
54         def _build_runner(self, idx_cfg: Dict[str, Any]) -> Tuple[DetectorRunner, Dict[str,
55         Any]]:
56             """Return (runner, detector-specific detection_params extras)."""
57             raise NotImplementedError
58
59         @staticmethod
60         def _probe_fps_width(video_path: str) -> tuple:
61             """Return (fps, frame_width) for a video, with sensible fallbacks."""
62             cap = cv2.VideoCapture(video_path)
```

```

61         fps = cap.get(cv2.CAP_PROP_FPS) if cap.isOpened() else 0.0
62         width = cap.get(cv2.CAP_PROP_FRAME_WIDTH) if cap.isOpened() else 0.0
63         cap.release()
64         return (fps if fps > 0 else 30.0, width if width > 0 else 1280.0)
65
66     @staticmethod
67     def _shot_count_from(segment: Dict[str, Any], duration: float) -> int:
68         """Provider-reported exchange count, else a duration-based estimate.
69
70         One canonical implementation ? the twins had drifted (wasb relied on
71         ``int(None)`` raising into the fallback; same result, by accident).
72         """
73         shot_count = segment.get("shuttle_exchange_count")
74         if shot_count is None:
75             return max(3, int(duration / 0.8))
76         try:
77             return int(shot_count)
78         except (ValueError, TypeError):
79             return max(3, int(duration / 0.8))
80
81     # --- Fusion seams (identity by default ? wasb_hybrid/tracknet_hybrid unchanged)
82     -----
83     def _enrich_candidate_stats(self, cw: Dict[str, Any], points: List[Any], fps: float,
84                                frame_width: float, video_path: str,
85                                idx_cfg: Dict[str, Any]) -> Dict[str, Any]:
86         """Stats dict persisted into ``candidate_segments.stats`` for one window. The
87         BASE
88         returns exactly the 4 motion keys, so the trajectory twi...
89         [/external_agent_tool_result]

```

26. assistant (2026-07-02T06:11:33.154Z)

Now I fully understand the served path. Critical detail: the **served** windowing uses production params (`velocity_thresh=0.02`, `merge_gap=2.0`, `min_window_duration=2.0` from config), which differ from the offline harness's recall-oriented proposal `(0.005, 1.0, 0.5)`. So the faithful served-side test must window with production params and then call the **real** `FusionSegmenter._enrich_candidate_stats/_select_for_handoff`. Let me check how the app loads config and instantiates segmenters so I reproduce production exactly, and confirm the environment (numpy/cv2/rally_gate).

27. assistant (2026-07-02T06:11:33.155Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\base.py
[/external_agent_tool_call]

```

28. assistant (2026-07-02T06:11:33.155Z)

```

[external_agent_tool_result]
1     from abc import ABC, abstractmethod
2     from typing import List, Dict, Any, Type, Optional
3     from backend.infrastructure.database import Database
4     from backend.results import Result # new standardized contract (Result[Value] ==
Success[Value] | Failure)
5
6     class BasePlaySegmenter(ABC):
7         def __init__(self, db: Database, config: Dict[str, Any]):
8             self.db = db
9             self.config = config
10
11         @property
12         @abstractmethod
13         def name(self) -> str:
14             """Unique identifier for this segmenter."""

```

```

15         pass
16
17     @property
18     @abstractmethod
19     def description(self) -> str:
20         """Human-readable explanation of this segmenter."""
21         pass
22
23     @abstractmethod
24     def process_video(self, video_id: str, video_path: str, player_pool: List[str] =
None, max_frames: int = None) -> "Result[List[Dict[str, Any]]]":
25         """
26         Processes a video to detect play segments, populates SQLite database,
27         and returns a Result.
28
29         Use isinstance(result, Failure) (or result.is_success()) to distinguish
30         "no segments found" from "something went wrong" (missing key, healthcheck
31         failure, etc.). This is the start of the standardized error/result contract.
32         """
33         pass
34
35     class SegmenterRegistry:
36     def __init__(self):
37         self._segmenters: Dict[str, Type[BasePlaySegmenter]] = {}
38
39     def register(self, segmenter_cls: Type[BasePlaySegmenter]):
40         """Registers a new segmenter class under its name property."""
41         try:
42             temp_inst = segmenter_cls(None, {})
43             name = temp_inst.name
44             except Exception as e:
45                 raise ValueError(f"Failed to resolve name property from segmenter class
{segmenter_cls.__name__}: {e}") from e
46             self._segmenters[name] = segmenter_cls
47
48     def get(self, name: str) -> Optional[Type[BasePlaySegmenter]]:
49         return self._segmenters.get(name)
50
51     def list_available(self) -> Dict[str, str]:
52         """Returns a dict of registered segmenter names and their descriptions."""
53         res = {}
54         for name, cls in self._segmenters.items():
55             try:
56                 temp_inst = cls(None, {})
57                 desc = temp_inst.description
58             except Exception:
59                 desc = "No description available."
60             res[name] = desc
61         return res
62
63     segmenter_registry = SegmenterRegistry()
64
[/external_agent_tool_result]

```

29. assistant (2026-07-02T06:11:33.155Z)

```

[external_agent_tool_call: Bash]
description: Find config loading and segmenter instantiation
command: echo "=== how config is loaded + segmenter instantiated in run_process_and_stitch ==="
&& grep -n "config\|segmenter\|registry\|get_config\|load_config\|model_dump\|\|.config"
run_process_and_stitch.py 2>/dev/null | head -40
[/external_agent_tool_call]

```

30. assistant (2026-07-02T06:11:33.155Z)

```
[external_agent_tool_result]
=== how config is loaded + segmenter instantiated in run_process_and_stitch ===
[/external_agent_tool_result]
```

31. assistant (2026-07-02T06:11:33.155Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\detectors\rally_gate.py
[/external_agent_tool_call]
```

32. assistant (2026-07-02T06:11:33.155Z)

```
[external_agent_tool_result]
1      """Rally-start gate: suppress between-point false fires using shuttle net-crossings.
2
3      WASB (shuttle-only) over-detects: when a player flicks/tosses the shuttle back to
4      the opponent for service after losing a point, that single trajectory looks like a
5      rally. The discriminator is structural, not duration-based: a real rally has the
6      shuttle crossing the net MULTIPLE times (each shot lands on the other side); a
7      single service feed crosses ~once.
8
9      This module is camera-agnostic and needs NO new model ? it works on the trajectory
10     we already produce (Frame,Visibility,X,Y). It estimates the play axis (the image
11     axis with the larger shuttle spread) and the net position (median shuttle coord on
12     that axis ? the shuttle clusters at the two ends and crosses the net region often,
13     so the median sits near the net), then counts sign changes of (coord - net) across
14     the visible points inside a candidate window, with a deadband to ignore jitter.
15
16     Gate A in the over-fire fix spectrum (B = player-stance state machine, future).
17     Pure functions only ? no I/O, unit-tested; intended as a post-filter on the
18     windows from trajectory_to_action_windows, or to fold into it later.
19     """
20     from __future__ import annotations
21
22     from dataclasses import dataclass
23     from typing import List, Optional, Sequence, Tuple
24
25     Interval = Tuple[float, float]
26
27
28     @dataclass
29     class GatedWindow:
30         start: float
31         end: float
32         crossings: int
33         visible_points: int
34         kept: bool
35
36
37     def _spread(vals: Sequence[float]) -> float:
38         """Robust spread = p95 - p5 (ignores a few outlier detections)."""
39         if len(vals) < 2:
40             return 0.0
41         s = sorted(vals)
42         lo = s[max(0, int(0.05 * (len(s) - 1)))]
43         hi = s[min(len(s) - 1, int(0.95 * (len(s) - 1)))]
44         return hi - lo
45
46
47     def _median(vals: Sequence[float]) -> float:
48         if not vals:
49             return 0.0
50         s = sorted(vals)
51         n = len(s)
```

```

52         return s[n // 2] if n % 2 else 0.5 * (s[n // 2 - 1] + s[n // 2])
53
54
55     def estimate_play_axis(points) -> Tuple[str, float, float]:
56         """Return (axis 'x'/'y', net_value, span) from visible points.
57
58         Play axis = the image axis along which the shuttle travels most (larger spread).
59         For a baseline camera that's Y (players top/bottom); for a side camera, X.
60         """
61         xs = [p.x for p in points if p.visible]
62         ys = [p.y for p in points if p.visible]
63         sx, sy = _spread(xs), _spread(ys)
64         if sx >= sy:
65             return "x", _median(xs), sx
66         return "y", _median(ys), sy
67
68
69     def count_net_crossings(window_points, axis: str, net: float, deadband: float) -> int:
70         """Count sign changes of (coord - net) across visible points, ignoring the
71         deadband zone near the net (so jitter at the net doesn't inflate the count)."""
72         crossings = 0
73         last_side = None
74         for p in window_points:
75             if not p.visible:
76                 continue
77             v = (p.x if axis == "x" else p.y) - net
78             if abs(v) < deadband:
79                 continue
80             side = v > 0
81             if last_side is not None and side != last_side:
82                 crossings += 1
83             last_side = side
84         return crossings
85
86
87     def _points_in(points, fps: float, start: float, end: float):
88         return [p for p in points if start * fps <= p.frame <= end * fps]
89
90
91     def gate_windows(points, windows: List[Interval], fps: float,
92                     min_crossings: int = 2, deadband_frac: float = 0.06,
93                     estimate: Optional[Tuple[str, float, float]] = None) ->
List[GatedWindow]:
94         """Annotate each (start,end) window with its shuttle net-crossing count and a
95         keep decision (crossings >= min_crossings). Deadband is a fraction of the play
96         axis span. Net/axis are estimated once over the whole trajectory (stable).
97
98         `estimate` lets a caller scoring MANY window...
[/external_agent_tool_result]

```

33. assistant (2026-07-02T06:11:33.155Z)

rally_gate is pure (no numpy needed). `FusionSegmenter(None, {})` is instantiable. The two seams only read the passed-in `idx\_cfg`. Now let me find the app's config loader so I reproduce production `idx\_cfg` exactly (not hand-build it), and confirm cached golden trajectories + the score/label tooling exist.

34. assistant (2026-07-02T06:11:33.155Z)

```

[external_agent_tool_call: Bash]
description: Find config loader, config keys, verify golden data present
command: echo "=== config loader ===" && ls backend/config/ && echo "---" && grep -rn "def
load_config\|def get_config\|json.load\|model_dump\|class AppConfig" backend/config/*.py | head
-30
echo; echo "=== top-level config.json keys ==="; python -c "import

```

```

json;print(list(json.load(open('config.json')).keys()))"
echo; echo "=== golden trajectories present? ==="; python - <<'PY'
import json, os
for s in json.load(open('output/human_lovo_manifest.json')):
    t=os.path.exists(s['trajectory']); l=os.path.exists(s['labels'])
    print(f"{s['name']:<22} traj={ 'OK' if t else 'MISSING' } labels={ 'OK' if l else 'MISSING' }
fps={s['fps']}")
PY
[/external_agent_tool_call]

```

35. assistant (2026-07-02T06:11:33.155Z)

```

[external_agent_tool_result]
=== config loader ===
__init__.py
__pycache__
loader.py
models.py
---
backend/config/loader.py:6:This replaces scattered json.load + .get() patterns throughout the
codebase.
backend/config/loader.py:50:     return AppConfig().model_dump(mode="json")
backend/config/loader.py:53:def load_config(path: str | Path | None = None) -> AppConfig:
backend/config/loader.py:69:         file_data = json.load(f)
backend/config/loader.py:100:     cfg_path.write_text(cfg.model_dump_json(indent=4),
encoding="utf-8")
backend/config/models.py:240:class AppConfig(BaseModel):
backend/config/models.py:269:     # via model_dump, so legacy chained access keeps working ?
returning the Pydantic
backend/config/models.py:289:         return val.model_dump(mode="json")

=== top-level config.json keys ===
['sport', 'ai_provider', 'ai_model', 'validation', 'indexing', 'output', 'stitching']

=== golden trajectories present? ===
adarsh_avi_singles      traj=OK labels=OK fps=59.94
kushagra_singles        traj=OK labels=OK fps=59.94
largetest_doubles        traj=OK labels=OK fps=29.97
gbaaddy                 traj=OK labels=OK fps=59.94
testlarge_short          traj=OK labels=OK fps=29.97
mahadevpura_singles      traj=OK labels=OK fps=29.97
[/external_agent_tool_result]

```

36. assistant (2026-07-02T06:11:33.155Z)

```

[external_agent_tool_call: Bash]
description: Check numpy/cv2 and eval helper signatures
command: echo "=== env: numpy/cv2/torch present? ===" && python -c "import numpy; print('numpy',
numpy.__version__)" 2>&1 | head -1; python -c "import cv2; print('cv2', cv2.__version__)" 2>&1 |
head -1
echo "=== load_golden_gts + score + label_candidates signatures ==="
grep -n "def load_golden_gts" backend/eval/calibrate_wasb.py
grep -n "def score\b|def score(" backend/eval/metrics.py
grep -n "def label_candidates" backend/eval/training.py
grep -n "def merge_overlaps" backend/eval/gemini_refine.py
[/external_agent_tool_call]

```

37. assistant (2026-07-02T06:11:33.155Z)

```

[external_agent_tool_result]
=== env: numpy/cv2/torch present? ===
numpy 2.4.6
cv2 4.13.0
=== load_golden_gts + score + label_candidates signatures ===

```

```

33: def load_golden_gts(csv_path: str) -> List[Interval]:
104: def score(preds: List[Interval], gts: List[Interval],
50: def label_candidates(candidate_intervals: List[Interval], gt_intervals: List[Interval],
39: def merge_overlaps(intervals: List[Interval], gap_tol: float = 0.0) -> List[Interval]:
[/external_agent_tool_result]

```

38. assistant (2026-07-02T06:11:33.155Z)

All data present (6/6 golden trajectories + labels), numpy/cv2 available. Let me read the config loader and the AppConfig dict-like access so I instantiate the segmenter exactly as production does.

39. assistant (2026-07-02T06:11:33.155Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\loader.py
[/external_agent_tool_call]

```

40. assistant (2026-07-02T06:11:33.155Z)

```

[external_agent_tool_result]
1      """Configuration loader and saver.
2
3      Provides a single place to load validated AppConfig with built-in defaults
4      merged with optional file overrides (config.json).
5
6      This replaces scattered json.load + .get() patterns throughout the codebase.
7      """
8
9      from __future__ import annotations
10
11     import json
12     import logging
13     from pathlib import Path
14     from typing import Any
15
16     from pydantic import BaseModel
17
18     from .models import AppConfig
19
20     logger = logging.getLogger(__name__)
21
22     DEFAULT_CONFIG_PATH = Path("config.json")
23
24
25     def _unknown_key_paths(data: dict[str, Any], model_cls: type[BaseModel], prefix: str =
""") -> list[str]:
26         """Dotted paths in `data` that the typed config will not honor.
27
28         Two silent failure modes exist without this check: a typo'd TOP-LEVEL key is
29         kept by AppConfig's ``extra="allow"`` but never read by typed code, and a
30         typo'd key INSIDE a section is silently dropped by Pydantic. Either way the
31         user's intent is lost ? collect every such key so load_config can warn.
32         """
33         unknown: list[str] = []
34         fields = model_cls.model_fields
35         for key, val in data.items():
36             if key not in fields:
37                 unknown.append(prefix + key)
38                 continue
39             ann = fields[key].annotation
40             if isinstance(val, dict) and isinstance(ann, type) and issubclass(ann,
BaseModel):
41                 unknown.extend(_unknown_key_paths(val, ann, f"{prefix}{key}.")

```

```

42         return unknown
43
44
45     def _get_builtin_defaults() -> dict[str, Any]:
46         """Return a plain dict of the current built-in defaults.
47
48         The AppConfig model is the single source of truth for defaults.
49         """
50         return AppConfig().model_dump(mode="json")
51
52
53     def load_config(path: str | Path | None = None) -> AppConfig:
54         """Load configuration with the following precedence:
55
56         1. Built-in defaults defined in the Pydantic models.
57         2. Values from the JSON file (if present and readable).
58         3. (Future) environment / CLI overrides.
59
60         Missing file or unreadable file ? returns a fully defaulted AppConfig.
61         Unknown keys in the file are tolerated (extra="allow" on the model).
62         """
63         cfg_path = Path(path) if path is not None else DEFAULT_CONFIG_PATH
64
65         file_data: dict[str, Any] = {}
66         if cfg_path.exists():
67             try:
68                 with open(cfg_path, "r", encoding="utf-8") as f:
69                     file_data = json.load(f)
70             except Exception as exc:
71                 # Non-fatal: continue with defaults + whatever we could parse.
72                 # In production we might log here.
73                 logger.warning(f"Failed to read {cfg_path}: {exc}. Using defaults + partial
data.")
74
75         if file_data:
76             unknown = _unknown_key_paths(file_data, AppConfig)
77             if unknown:
78                 logger.warning(
79                     "[CONFIG WARNING] %s contains keys the typed config does not "
80                     "recognize (typo'd or removed knobs ? top-level extras are kept "
81                     "but unread; unknown section keys are silently dropped): %s",
82                     cfg_path, " ".join(sorted(unknown)),
83                 )
84
85         # Merge: start from defaults, overlay file data (Pydantic will validate/coerce).
86         merged = {**_get_builtin_defaults(), **file_data}
87
88         return AppConfig.model_validate(merged)
89
90
91     def save_default_config(path: str | Path | None = None) -> Path:
92         """Write a fresh default config.json.
93
94         Used by setup.py diagnostics and the --setup CLI path.
95         Overwrites if the file exists.
96         """
97         cfg_path = Path(path) if path is not None else DEFAULT_CONFIG_PATH
98         cfg = AppConfig()
99         cfg_path.parent.mkdir(parents=True, exist_ok=True)
100         cfg_path.write_text(cfg.model_dump_json(indent=4), encoding="utf-8")
101         return cfg_path
102
103
104     # Backwards-compatible alias u...
[/external_agent_tool_result]

```

41. assistant (2026-07-02T06:11:33.155Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\models.py
[/external_agent_tool_call]
```

42. assistant (2026-07-02T06:11:33.156Z)

```
[external_agent_tool_result]
240     class AppConfig(BaseModel):
241         """Root configuration object.
242
243         All runtime configuration should be accessed via an instance of this class
244         (or the loader) rather than raw dict lookups.
245         """
246
247         sport: Sport = "badminton"
248         ai_provider: str = "gemini"
249         # Auto-tracking alias: 2.5-flash is legacy now that 3.5-flash is the stable
flagship;
250         # the alias never deprecates underneath us. Eval reports record the run date
alongside.
251         ai_model: str = "gemini-flash-latest"
252
253         validation: ValidationConfig = Field(default_factory=ValidationConfig)
254         indexing: IndexingConfig = Field(default_factory=IndexingConfig)
255         output: OutputConfig = Field(default_factory=OutputConfig)
256         stitching: StitchingConfig = Field(default_factory=StitchingConfig)
257         security: SecurityConfig = Field(default_factory=SecurityConfig)
258         logging: LoggingConfig = Field(default_factory=LoggingConfig)
259
260         # Allow extra keys during transition so existing config.json files do not break.
261         model_config = {
262             "extra": "allow",
263             "validate_assignment": True,
264         }
265
266         # Temporary migration shim for legacy dict-style access (config.get("section",
{ }).get(...)).
267         # Will be removed once all call sites (validators, segmenters, etc.) move to typed
268         # attribute access / the typed config. A nested *section* is returned as a plain
dict
269         # via model_dump, so legacy chained access keeps working ? returning the Pydantic
270         # section model instead breaks it (a model has no .get), which bit every validator
on
271         # real runs. Scalar leaves are returned as-is.
272         def get(self, key: str, default: Any = None) -> Any:
273             _MISSING = object()
274             val: Any = _MISSING
275             if key in type(self).model_fields or hasattr(self, key):
276                 val = getattr(self, key, _MISSING)
277             if val is _MISSING:
278                 # Legacy convenience: find a leaf key inside any nested section. Sections
are
279                 # discovered dynamically from the model fields, so a future top-level
section
280                 # (e.g. `audio`, `jobs`) is covered automatically without editing a
hardcoded list.
281                 for fname in type(self).model_fields:
282                     section = getattr(self, fname, None)
283                     if isinstance(section, BaseModel) and hasattr(section, key):
284                         val = getattr(section, key)
285                         break
286             if val is _MISSING:
287                 return default
```

```

288         if isinstance(val, BaseModel):
289             return val.model_dump(mode="json")
290         return val
291
292     def __getitem__(self, key: str) -> Any:
293         val = self.get(key)
294         if val is None and key not in type(self).model_fields:
295             # Re-raise like a real dict for unknown top-level
296             raise KeyError(key)
297         return val
298
299
300     # Convenience type alias for callers that want to annotate
301     Config = AppConfig
302
303 [/external_agent_tool_result]

```

43. assistant (2026-07-02T06:11:33.156Z)

Everything needed is present and I now know how to reproduce production exactly: ``load_config()`` ? ``config.get("indexing")`` is a plain dict, and the segmenter seems only read the passed-in ``idx_cfg``. Let me confirm the runner's parse/windowing signatures and the ``score`` function so my harness calls them correctly.

44. assistant (2026-07-02T06:11:33.156Z)

```

[external_agent_tool_call: Bash]
description: Inspect runner parsing/windowing and scoring signatures
command: echo "=== parse_trajectory_csv + trajectory_to_action_windows signatures ===" && grep
-n "def parse_trajectory_csv\|def trajectory_to_action_windows"
backend/pipeline/detectors/tracknet_runner.py
echo; echo "=== trajectory_to_action_windows full signature ==="; sed -n '/def
trajectory_to_action_windows/,/min_window_duration/p'
backend/pipeline/detectors/tracknet_runner.py | head -20
echo; echo "=== metrics.score ==="; sed -n '104,150p' backend/eval/metrics.py
echo; echo "=== label_candidates ==="; sed -n '50,90p' backend/eval/training.py
[/external_agent_tool_call]

```

45. assistant (2026-07-02T06:11:33.156Z)

```

[external_agent_tool_result]
=== parse_trajectory_csv + trajectory_to_action_windows signatures ===
233:     def parse_trajectory_csv(csv_win_path: str) -> List[TrajectoryPoint]:
252:     def trajectory_to_action_windows(

=== trajectory_to_action_windows full signature ===
    def trajectory_to_action_windows(
        points: List[TrajectoryPoint],
        fps: float,
        frame_width: float,
        chunk_start: float = 0.0,
        velocity_thresh: float = 0.02,
        merge_gap: float = 2.0,
        min_window_duration: float = 2.0,

=== metrics.score ===
def score(preds: List[Interval], gts: List[Interval],
        iou_threshold: float = 0.5,
        match_result: Optional[MatchResult] = None) -> Score:
    """Compute precision/recall/F1, mean matched IoU, and boundary errors.

    Pass a precomputed `match_result` to avoid re-matching (e.g. when also
    extracting the failure lists); otherwise matching is done here.
    """

```

```

    mr = match_result if match_result is not None else match_intervals(preds, gts,
iou_threshold)

    tp = len(mr.matches)
    fp = len(mr.unmatched_pred_indices)
    fn = len(mr.unmatched_gt_indices)

    precision = tp / (tp + fp) if (tp + fp) > 0 else 0.0
    recall = tp / (tp + fn) if (tp + fn) > 0 else 0.0
    f1 = (2 * precision * recall / (precision + recall)) if (precision + recall) > 0 else 0.0

    if mr.matches:
        mean_iou = sum(m.iou for m in mr.matches) / tp
        mean_start_err = sum(abs(preds[m.pred_index][0] - gts[m.gt_index][0]) for m in
mr.matches) / tp
        mean_end_err = sum(abs(preds[m.pred_index][1] - gts[m.gt_index][1]) for m in mr.matches)
/ tp
    else:
        mean_iou = mean_start_err = mean_end_err = 0.0

    return Score(
        iou_threshold=iou_threshold,
        num_pred=len(preds), num_gt=len(gts),
        true_positives=tp, false_positives=fp, false_negatives=fn,
        precision=precision, recall=recall, f1=f1,
        mean_iou=mean_iou, mean_start_error=mean_start_err, mean_end_error=mean_end_err,
    )

```

```

def pr_curve(preds: List[Interval], gts: List[Interval],
            thresholds: Optional[List[float]] = None) -> List[Score]:
    """Score across a sweep of IoU thresholds (default 0.1..0.9 step 0.1).

    Lets you see strict-vs-lenient behaviour rather than a single operating point.
    """
    if thresholds is None:
        thresholds = [round(0.1 * i, 1) for i in range(1, 10)]
    return [score(preds, gts, iou_threshold=t) for t in thresholds]

```

```

=== label_candidates ===
def label_candidates(candidate_intervals: List[Interval], gt_intervals: List[Interval],
                    iou_threshold: float = 0.5) -> List[int]:
    """Label each candidate window 1 (real rally) or 0 (spurious).

    A window is positive iff it is matched one-to-one to a ground-truth rally at
    `iou_threshold` ? the *same* matching the evaluator uses for true positives,
    so the classifier is trained on exactly what we score. Substrate-agnostic:
    works for YOLO or WASB-derived windows alike.
    """
    mr = match_intervals(candidate_intervals, gt_intervals, iou_threshold)
    positive = {m.pred_index for m in mr.matches}
    return [1 if i in positive else 0 for i in range(len(candidate_intervals))]

```

```

def build_training_set(
    candidate_rows: List[Dict],
    gt_intervals: List[Interval],
    iou_threshold: float = 0.5,
    feature_fn: Callable[[Dict], List[float]] = extract_yolo_features,
) -> Tuple[List[List[float]], List[int], List[Interval]]:
    """Turn one video's candidate rows + ground truth into (X, y, intervals).

    `feature_fn` is pluggable so a WASB substrate can supply shuttle-motion
    features without changing the labeling logic. Returns the feature matrix `X`,
    binary labels `y`, and the candidate intervals (handy for downstream merging).

```

```

"""
intervals: List[Interval] = [
    (float(r["start_time"]), float(r["end_time"])) for r in candidate_rows
]
y = label_candidates(intervals, gt_intervals, iou_threshold)
X = [feature_fn(r) for r in candi...
[/external_agent_tool_result]

```

46. assistant (2026-07-02T06:11:33.156Z)

I have everything. Let me confirm `wasb\_hybrid` inherits the base seams unchanged (so the mc=0 identity claim is real), then write the verification harness.

47. assistant (2026-07-02T06:11:33.156Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\wasb_hybrid.py
[/external_agent_tool_call]

```

48. assistant (2026-07-02T06:11:33.156Z)

```

[external_agent_tool_result]
1     """WASB-based hybrid play segmenter.
2
3     Phase 2 uses **WASB** (nttcom/WASB-SBDT, MIT) shuttle-trajectory detection via
4     WSL2 (backend/pipeline/detectors/wasb_runner.py). WASB ships pretrained
5     badminton weights and runs today, so this is the live shuttle-motion substrate.
6
7     For the per-video *calibration* goal we mainly use Phase 2 (trajectory ->
8     candidate windows); the windowing thresholds (velocity_thresh / merge_gap /
9     min_window_duration) are what get tuned against golden rally labels.
10
11     The pipeline engine (healthcheck -> trajectory -> windows -> AI handoff -> DB)
12     lives in trajectory_hybrid.TrajectoryHybridSegmenter; this class only wires the
13     WASB runner and identity.
14     """
15     from typing import Any, Dict, Tuple
16
17     from backend.pipeline.segmenters.base import segmenter_registry
18     from backend.pipeline.segmenters.trajectory_hybrid import TrajectoryHybridSegmenter
19     from backend.pipeline.detectors.base import DetectorRunner
20     from backend.pipeline.detectors.wasb_runner import WasbRunner, WasbConfig
21
22
23     class WasbHybridSegmenter(TrajectoryHybridSegmenter):
24         family = "wasb"
25         display_label = "WASB"
26
27         @property
28         def name(self) -> str:
29             return "wasb_hybrid"
30
31         @property
32         def description(self) -> str:
33             return ("Two-stage pipeline: WASB shuttle-trajectory detection (WSL2, pretrained
badminton "
34                     "weights) for precise candidate rallies + AI provider for semantic
boundaries.")
35
36         def _build_runner(self, idx_cfg: Dict[str, Any]) -> Tuple[DetectorRunner, Dict[str,
Any]]:
37             cfg = WasbConfig.from_indexing_cfg(idx_cfg)
38             return WasbRunner(cfg), {"weights_path": cfg.weights_path}

```

```
39
40
41     segmenter_registry.register(WasbHybridSegmenter)
42
[/external_agent_tool_result]
```

49. assistant (2026-07-02T06:11:33.156Z)

Confirmed: `WasbHybridSegmenter` inherits both base seams unchanged, and at `min\_crossings=0` the fusion windowing params equal wasb's ? so the mc=0 identity is real and I'll demonstrate it empirically rather than assert it. Now I'll write the served-side verification harness.

50. assistant (2026-07-02T06:11:33.156Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\served_gateA_verify.py
[/external_agent_tool_call]
```

51. assistant (2026-07-02T06:11:33.156Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\served_gateA_verify.py (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

52. assistant (2026-07-02T06:11:33.156Z)

Now let me run the served-side verification harness.

53. assistant (2026-07-02T06:11:33.156Z)

```
[external_agent_tool_call: Bash]
description: Run served-side Gate-A verification
command: python -m scratch.served_gateA_verify 2>&1 | tail -60
[/external_agent_tool_call]
```

54. assistant (2026-07-02T06:11:33.156Z)

```
[external_agent_tool_result]
=====
SERVED-SIDE Gate-A verification ? live FusionSegmenter over CACHED golden trajectories
(config.indexing.fusion = {'substrate': 'wasb', 'velocity_threshold': 0.02, 'cue_flags': {},
'min_crossings': 2, 'min_players': 0, 'court_polygon': None, 'net_axis': 'y', 'net_position':
0.5})
=====

video          win  mc0  mc2  drop  lost  F1@mc0  F1@mc2  ?F1  seed=wasb
handoff==wasb@mc0
adarsh_avi_singles    97   97   76   21    7   0.632   0.628 -0.004   True   True
kushagra_singles      2    2    2    0    0   0.000   0.000 +0.000   True   True
largetest_doubles     52   52   42   10    5   0.535   0.484 -0.051   True   True
gbaaddy              22   22   20    2    0   0.086   0.088 +0.003   True   True
testlarge_short        5    5    5    0    0   0.182   0.182 +0.000   True   True
mahadevpura_singles   24   24   23    1    0   0.364   0.370 +0.007   True   True
-----
MEAN over n=6:  P 0.317->0.337  R 0.290->0.261  F1 0.300->0.292  (? -0.008)

(a) mc=0 seeding+handoff byte-identical to wasb_hybrid on all videos : YES ?
(b) real-rally windows dropped by Gate-A (mc=2) across corpus          : 12 ?
    F1 regression on any video                                         : YES ?

wrote output/gateA_served_verify/served_gateA_summary.json
[/external_agent_tool_result]
```

55. assistant (2026-07-02T06:11:33.156Z)

Important ? this is a substantive finding, not a rubber-stamp. Let me read it carefully:

- ***(a) holds****: at `mc=0` the fusion seeding + handoff is byte-identical to `wasb\_hybrid` on all 6 videos. ? The no-regression invariant on the real served path is confirmed.
- ***(b) does NOT reproduce the offline win****: on the ****production**** windowing, Gate-A drops ****12** windows that actually match golden rallies** (real-rally loss), mean F1 goes 0.300?0.292 (**?0.008**), with targetest_doubles ?0.051.

The offline `+0.074` was measured on the recall-oriented **proposal** windowing `(0.005,1.0,0.5)`; production serves a coarse windowing `(0.02,2.0,2.0)`. I must verify this is a real windowing-driven effect and not a harness artifact before recording it. Let me re-run the offline LOVO baseline and inspect the dropped real-rally windows.

56. assistant (2026-07-02T06:11:33.156Z)

```
[external_agent_tool_call: Bash]
description: Re-run offline LOVO to confirm proposal-windowing Gate-A number
command: echo "=== Re-run offline ab_n6_lovo (proposal windowing) to confirm +0.074 baseline
=== " && python -m scratch.ab_n6_lovo 2>&1 | sed -n '1,40p'
[/external_agent_tool_call]
```

57. assistant (2026-07-02T06:11:33.156Z)

```
[external_agent_tool_call: Bash]
description: Inspect net_crossings of dropped real-rally windows
command: echo "=== Dropped real-rally windows (net_crossings) on production windowing ===" &&
python - <<'PY'
import json
s=json.load(open('output/gateA_served_verify/served_gateA_summary.json'))
for v in s['per_video']:
    lost=[d for d in v['drops'] if d['is_real_rally']]
    if lost:
        nc=[d['net_crossings'] for d in lost]
        print(f"{v['name']:<22} lost={len(lost):>2} net_crossings of lost real rallies: {nc}")
        # also distribution of ALL drops
        fp=[d for d in v['drops'] if not d['is_real_rally']]
        print(f"  {'':<20} dropped-FPs={len(fp)} (correct drops) total-windows={v['num_windows']}")
gts={v['num_gts']}")
PY
[/external_agent_tool_call]
```

58. assistant (2026-07-02T06:11:33.156Z)

```
[external_agent_tool_result]
=== Re-run offline ab_n6_lovo (proposal windowing) to confirm +0.074 baseline ===
```

```
=====
FIRST A/B EXPERIMENT ? n=6 golden corpus (offline, persisted trajectories)
```

```
=====
adarsh_avi_singles      172 candidates   96 rallies   proposal-recall-ceiling=0.833
kushagra_singles       20 candidates   32 rallies   proposal-recall-ceiling=0.125
targetest_doubles      86 candidates   49 rallies   proposal-recall-ceiling=0.816
gbaaddy                82 candidates   48 rallies   proposal-recall-ceiling=0.312
testlarge_short        14 candidates    6 rallies   proposal-recall-ceiling=0.500
mahadevpura_singles    57 candidates   31 rallies   proposal-recall-ceiling=0.645
```

```
--- per-video held-out F1 ---
```

video	CONTROL	GateA	learned
adarsh_avi_singles	0.597	0.722	0.613
kushagra_singles	0.154	0.163	0.163
targetest_doubles	0.593	0.667	0.535
gbaaddy	0.231	0.274	0.222
testlarge_short	0.300	0.375	0.000

```

mahadevpura_singles      0.455      0.571      0.308

--- A/B 1: CONTROL vs Gate-A (static) ? C1 honest aggregation ---
  meanF1 0.388 -> 0.462 ?F1=+0.074 CI[0.042, 0.104] excl0=True sign={'wins': 6, 'losses': 0,
'ties': 0, 'n_effective': 6, 'p_value': 0.031}
--- A/B 2: CONTROL vs learned fusion (LOVO) ? C1 honest aggregation ---
  meanF1 0.388 -> 0.307 ?F1=-0.081 CI[-0.173, -0.002] excl0=True sign={'wins': 2, 'losses': 4,
'ties': 0, 'n_effective': 6, 'p_value': 0.688}

--- C4 over-segmentation-aware metrics (mean across videos) ---
  CONTROL   {'f1@10': 0.645, 'f1@25': 0.603, 'f1@50': 0.388, 'seg_count_ratio': 1.68}
  gateA     {'f1@10': 0.748, 'f1@25': 0.698, 'f1@50': 0.462, 'seg_count_ratio': 1.01}
  learned   {'f1@10': 0.564, 'f1@25': 0.509, 'f1@50': 0.307, 'seg_count_ratio': 0.89}
  learned out-of-fold (C3) mAP@tIoU[.3-.7] = 0.245

```

```

wrote output/ab_experiment_n6.json
[/external_agent_tool_result]

```

59. assistant (2026-07-02T06:11:33.157Z)

```

[external_agent_tool_result]
=== Dropped real-rally windows (net_crossings) on production windowing ===
adarsh_avi_singles      lost= 7 net_crossings of lost real rallies: [1.0, 0.0, 1.0, 1.0, 1.0,
1.0, 1.0]

                        dropped-FPs=14 (correct drops) total-windows=97 gts=96
                        dropped-FPs=0 (correct drops) total-windows=2 gts=32
largetest_doubles       lost= 5 net_crossings of lost real rallies: [1.0, 1.0, 1.0, 1.0, 1.0]
                        dropped-FPs=5 (correct drops) total-windows=52 gts=49
                        dropped-FPs=2 (correct drops) total-windows=22 gts=48
                        dropped-FPs=0 (correct drops) total-windows=5 gts=6
                        dropped-FPs=1 (correct drops) total-windows=24 gts=31
[/external_agent_tool_result]

```

60. assistant (2026-07-02T06:11:33.157Z)

Now the picture is rigorous and clear. Let me state what the two runs establish:

```

- **Offline harness reproduced exactly**: proposal windowing `(0.005,1.0,0.5)`, CONTROL F1 0.388
? Gate-A 0.462, ++0.074++, 6/6, p=0.031, seg_count_ratio **1.68?1.01** (it was cleaning up
heavy over-segmentation).
- **Served path**: production windowing `(0.02,2.0,2.0)` already merges aggressively (low over-
segmentation), so Gate-A has little FP to remove and instead **drops 12 real-rally windows whose
net_crossings are 0?1** (almost all exactly 1) ? mean F1 **?0.008**, worst on doubles.

```

The lost rallies sit at `net\_crossings=1`, which suggests `min\_crossings=1` might be the right served operating point. Let me sweep `min\_crossings ? {0,1,2,3}` on the production windowing to give the owner the operating-point curve ? that's directly decision-relevant.

61. assistant (2026-07-02T06:11:33.157Z)

```

[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\served_gateA_sweep.py
[/external_agent_tool_call]

```

62. assistant (2026-07-02T06:11:33.157Z)

```

[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\badminton-highlight-
indexer\scratch\served_gateA_sweep.py (file state is current in your context ? no need to Read
it back)
[/external_agent_tool_result]

```

63. assistant (2026-07-02T06:11:33.157Z)

```
[external_agent_tool_call: Bash]
description: Run Gate-A operating-point sweep on production windowing
command: python -m scratch.served_gateA_sweep 2>&1 | tail -25
[/external_agent_tool_call]
```

64. assistant (2026-07-02T06:11:33.157Z)

```
[external_agent_tool_result]
=====
Gate-A operating-point sweep on the SERVED (production) windowing (vt=0.02, merge_gap=2.0,
min_win=2.0)
=====
metric                mc=0          mc=1          mc=2          mc=3
mean F1                0.300        0.303        0.292        0.269
mean precision         0.317        0.327        0.337        0.338
mean recall            0.290        0.288        0.261        0.226
real rallies lost      0           1           12          28
?F1 vs mc=0           +0.000       +0.003       -0.008       -0.031

(mc=2 is the current served default from PR #146; mc=0 = Gate-A OFF = wasb_hybrid)
[/external_agent_tool_result]
```

65. assistant (2026-07-02T06:11:33.157Z)

The operating-point curve is decisive and decision-relevant. On the ****served**** (production) windowing:

```
| | mc=0 (off) | mc=1 | **mc=2 (shipped)** | mc=3 |
|---|---|---|---|
| mean F1 | 0.300 | **0.303** | 0.292 | 0.269 |
| precision | 0.317 | 0.327 | 0.337 | 0.338 |
| recall | 0.290 | 0.288 | 0.261 | 0.226 |
| real rallies lost | 0 | 1 | 12 | 28 |
| ?F1 vs off | ? | +0.003 | **?0.008** | ?0.031 |
```

So at the served windowing, `mc=2` is mildly net-negative and drops 12 real rallies; `mc=1` would be the better operating point if Gate-A is kept on. Let me record this honestly in `docs/QUALITY\_ITERATIONS.md`. Let me read its structure first.

66. assistant (2026-07-02T06:11:33.157Z)

```
[external_agent_tool_call: Bash]
description: Inspect QUALITY_ITERATIONS.md structure
command: echo "=== file size + headings ===" && grep -n "^#\|^##\|^### \|Gate-A\|Gate
A\|net_crossings\|I[0-9]\b\|^## I" docs/QUALITY_ITERATIONS.md | head -60; echo "---- total lines
----"; wc -l docs/QUALITY_ITERATIONS.md
[/external_agent_tool_call]
```

67. assistant (2026-07-02T06:11:33.157Z)

```
[external_agent_tool_result]
=== file size + headings ===
1:# Rally-detection quality iterations ? living log
25:## Baselines that drive strategy (IoU 0.5 vs human golden labels)
41:## Current track (live) ? newest first
43:### PERSONALIZATION Phase-0 quality checkpoint ? the eval/safety RAILS for the per-user
feature *(2026-06-14)*
52: promote noise and can't disable a structural guard (the proven Gate-A held-shuttle guard
stays on regardless of a thin
70:**Gate-A-only structural + `min_n = 5`*.
72:### Fusion P3 + P4 MEASURED on the n=6 golden corpus ? person = no lift, audio = promising
(not promotable) *(2026-06-14)*
91:### Gate-A flipped to the SERVED DEFAULT (owner-approved) *(2026-06-14)*
92:**Gate-A (net-crossings ? 2) is now the DEFAULT detection behaviour.** The owner approved
```

promoting the

```

95:- `backend/config/models.py` ? `FusionConfig.min_crossings` default **0 ? 2** (Gate A on
wherever
103:Gate A (ungated handoff, reproduces the substrate byte-for-byte), or
`indexing.default_segmenter: wasb_hybrid`
116:### Acting on the first A/B: Gate-A PROMOTED + learned variant FIXED + GX010125
corroboration *(2026-06-14)*
117:**Gate-A (net-crossings ? 2) ? PROMOTED (first cue promoted via the harness).** The measured
win
120:bar. **Recommended config:** `indexing.fusion.min_crossings = 2` on the fusion path (the
eval-only Gate A
125:CONTROL 45 rallies vs **Gate-A 27**; Gate-A dropped **18 low-crossing windows (~1.7 min)**
of likely
135:**Next:** a real fusion-path run to confirm Gate-A served-side; per-video output layout
138:### First end-to-end A/B on the golden corpus (n=6) ? **ARCHIVED** *(2026-06-14)*
140:**Gate-A (net-crossings ? 2) is a measured win** ? ?F1 +0.074, 6/6 matches, sign-test **p =
0.031**, CI
146:Next: Gate-A is a candidate for the no-regression promotion gate; the logistic failure is
the C2 (calibration)
149:### Eval-honesty helpers (course corrections C1?C4) ? additive, default-OFF *(2026-06-14)*
173:### Multi-signal fusion P2: the FusionSegmenter *(in progress, 2026-06-14)*
174:**Hypothesis.** Adjudicating shuttle-seeded windows with the net-crossing **Gate A**
(`min_crossings?2`)
188:**Data support for Gate A.** The same archive shows real rallies on the multi-court matches
contain
192:**Shipping shape.** A NEW `fusion` segmenter, **default OFF** (`default_segmenter`
unchanged); Gate A
210:Gate-A-pruned windows (a P3 redesign, would change the persisted-superset semantics); and
`fusion.velocity_threshold`
215:### Person cue extracted to a reusable producer (fusion P1) ?
[#115](https://github.com/avidullu/badminton-highlight-indexer/pull/115), 2026-06-13
221:### Experiment harness: A/B without rollback risk ?
[#114](https://github.com/avidullu/badminton-highlight-indexer/pull/114), 2026-06-13
231:## Archived predecessors (the historical trail)
245:## Standing lessons (carry-forward)
258: Gate A leans on this.
264:- **Gate A already existed, just wasn't wired to production** ? the cheapest precision win
is plumbing, not
---- total lines ----
265 docs/QUALITY_ITERATIONS.md
[/external_agent_tool_result]

```

68. assistant (2026-07-02T06:11:33.157Z)

Let me read the existing Gate-A entries to match style and place my entry correctly.

69. assistant (2026-07-02T06:11:33.157Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\QUALITY_ITERATIONS.md
[/external_agent_tool_call]

```

70. assistant (2026-07-02T06:11:33.157Z)

```

[external_agent_tool_result]
41     ## Current track (live) ? newest first
42
43     ### PERSONALIZATION Phase-0 quality checkpoint ? the eval/safety RAILS for the per-user
feature *(2026-06-14)*
44     The owner adopted the **HYBRID** posture: personalization as a **flag-gated FEATURE on a
generalization-first baseline**,
45     paired with a **contribution flywheel** so per-user gains and the shared baseline
compound together (full design in
46     [PERSONALIZATION_PLAN.md](PERSONALIZATION_PLAN.md), merged

```

```

[#144](https://github.com/avidullu/badminton-highlight-indexer/pull/144)).
47     This checkpoint landed the three **honest-eval RAILS** that keep a per-user/small-N
tuning layer from promoting noise or
48     silently regressing a user:
49
50     - **Per-user promotion gate ? [#141](https://github.com/avidullu/badminton-highlight-
indexer/pull/141)**
51     (`backend/eval/per_user_gate.py`): a **min-N floor** + **structural-guard exemption**
? per-user/small-N tuning can't
52     promote noise and can't disable a structural guard (the proven Gate-A held-shuttle
guard stays on regardless of a thin
53     user corpus).
54     - **Held-out-USER learning curve ? [#142](https://github.com/avidullu/badminton-
highlight-indexer/pull/142)**
55     (`backend/eval/per_user_eval.py`): turns the vague "<K videos to superior quality"
claim into a **falsifiable release
56     number** (the `smallest_k_with_lift` measured on held-out users).
57     - **Per-user no-regression guard ? [#143](https://github.com/avidullu/badminton-
highlight-indexer/pull/143)**: a
58     retrain/baseline-upgrade can **never make a user worse than their incumbent** (frozen
per-user CONTROL; a regressing
59     candidate is refused).
60
61     **Read HONESTLY:** these are the **eval/safety rails** for the personalization feature ?
**not** new measured
62     rally-detection wins. No F1 moved this checkpoint; what landed is the discipline that
lets per-user tuning ship without
63     regressing the baseline or over-claiming on n=6-per-user data.
64
65     **Locked owner decisions (2026-06-14):** identity = *design for the north-star (multi-
user/cloud), implement for current
66     use (single-user local)*; the inline per-user classifier sits **behind a cloud-load
interface** (local store = one impl);
67     **`min_n` is a PROMOTION-confidence floor, NOT a service/segmentation gate** (a one-
video user is ALWAYS fully segmented by
68     the baseline batteries); **free-tier videos ARE pooled** into the shared baseline
(consent baked into free terms,
69     minors-exclusion enforced at the pooling query); and the previously-pending structural-
guard sub-decision is now LOCKED ?
70     **Gate-A-only structural + `min_n = 5`**.
71
72     ### Fusion P3 + P4 MEASURED on the n=6 golden corpus ? person = no lift, audio =
promising (not promotable) *(2026-06-14)*
73     The GPU person-feature extraction finished all 6 golden videos (`fusion_golden.py` ?
`output/*_golden_features.json`, the
74     replayable substrate); the CPU LOVO drivers then measured the lift with **honest stats
(C1 ? bootstrap 95% CI + paired sign
75     test; never a bare mean at n=6)**:
76
77     | Variant (honest LOVO, IoU ? 0.5) | F1 | ?F1 vs prior | 95% CI | sign test | CI clears
0 |
78     | ---|---|---|---|---|---|
79     | shuttle-only | 0.307 | ? | ? | ? | ? |
80     | + person (P3) | 0.286 | **?0.021** (?6.7%) | [?0.165, +0.162] | 2W/4L (p=0.688) |
**No** |
81     | + person + audio (P4 incr.) | 0.366 | **+0.079** (+27.7%) | [?0.054, +0.203] | 4W/2L
(p=0.688) | **No** |
82
83     **Read honestly:** neither cue clears the promotion bar (both CIs span 0, both sign
tests n.s.) at n=6 ? **both stay
84     default-OFF** (promote-only-on-lift). **Person** adds noise (no court polygons ?
`players_in_court` counts spectators);
85     per-video it's a wash (testlarge +0.40, gbaaddy ?0.22). **Audio** is the more promising
lead ? +0.079 / +27.7%, wins 4/6
86     (largetest +0.27, mahadevpura +0.27), and shuttle+person+audio (0.366) lands **above**

```

```
shuttle-only (0.307) ? but n=6 can't
87      confirm it. **Next:** more matches (raise N until the sign test can resolve ~...
[/external_agent_tool_result]
```

71. assistant (2026-07-02T06:11:33.157Z)

The existing flip entry (lines 110?114) explicitly anticipated this exact follow-up: `"the offline ?F1 is on the recall-oriented proposal candidate set? a golden served regression on the fusion path remains a worthwhile optional follow-up."` My run answers it. Let me record the honest result at the top of the live track.

72. assistant (2026-07-02T06:11:33.157Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\QUALITY_ITERATIONS.md
[/external_agent_tool_call]
```

73. assistant (2026-07-02T06:11:33.157Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\QUALITY_ITERATIONS.md has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

74. assistant (2026-07-02T06:11:33.157Z)

Now let me update the live memory checkpoint with this finding.

75. assistant (2026-07-02T06:11:33.158Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md
[/external_agent_tool_call]
```

76. assistant (2026-07-02T06:11:33.158Z)

```
[external_agent_tool_result: error]
<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>
[/external_agent_tool_result]
```

77. assistant (2026-07-02T06:11:33.158Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md
[/external_agent_tool_call]
```

78. assistant (2026-07-02T06:11:33.158Z)

```
[external_agent_tool_result]
1      ---
2      name: current-state
3      description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
only on request). What just landed, what's next, open decisions."
4      metadata:
5        node_type: memory
6        type: project
7        originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8      ---
9
10     # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
```

12 **Checkpoint:** 2026-06-14 (PERSONALIZATION Phase-0 LANDED + quality checkpoint).
Owner adopted the HYBRID ? personalization as a
[/external_agent_tool_result]

79. assistant (2026-07-02T06:11:33.158Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md
[/external_agent_tool_call]

80. assistant (2026-07-02T06:11:33.158Z)

[external_agent_tool_result]
The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

81. assistant (2026-07-02T06:11:33.158Z)

Recorded. Let me verify the working-tree state (I have not touched `config.json` ? the rollback is owner-gated).

82. assistant (2026-07-02T06:11:33.158Z)

[external_agent_tool_call: Bash]
description: Confirm working tree state and config untouched
command: git status --short && echo "--- confirm config.json untouched (still mc=2) ---" && git diff --stat config.json && python -c "import json;print('min_crossings =', json.load(open('config.json'))['indexing']['fusion']['min_crossings'])"
[/external_agent_tool_call]

83. assistant (2026-07-02T06:11:33.158Z)

[external_agent_tool_result]
M docs/QUALITY_ITERATIONS.md
?? artifacts/
--- confirm config.json untouched (still mc=2) ---
min_crossings = 2
[/external_agent_tool_result]

84. assistant (2026-07-02T06:11:33.158Z)

Verification complete and recorded. Only `docs/QUALITY\_ITERATIONS.md` is modified; `config.json` is untouched (still `mc=2`), scratch scripts and artifacts are in gitignored dirs. Here's what I found.

Served-side Gate-A verification ? result

I drove the **live** `FusionSegmenter` served seams (`_enrich_candidate_stats` ? `net_crossings`, `select_for_handoff` ? the gate) over each golden video's cached WASB trajectory, at the production windowing `process_video` actually uses (vt=0.02, merge_gap=2.0, min_win=2.0`, loaded via load_config()). No WASB re-run, no GPU, no Gemini. Repro: [served_gateA_verify.py](scratch/served_gateA_verify.py) + [served_gateA_sweep.py](scratch/served_gateA_sweep.py).`

(a) No-regression invariant ? **CONFIRMED ?**. At `min_crossings=0``, `fusion_hybrid`'s` seeding **and** handoff set are **byte-identical** to `wasb_hybrid`` on all 6 golden videos. The "fusion adds nothing when the gates are off" property is now verified on the **live** served code, not just by #117's review.

(b) The `+0.074`` win does NOT transfer to the served path `??`. That win was measured on the

eval harness's recall-oriented **proposal** windowing `(0.005,1.0,0.5)`, where CONTROL over-segments (seg-ratio 1.68) and Gate-A's job is cleaning up fragmentation (?1.01). Production serves a **coarse** windowing that already merges aggressively ? so Gate-A has little FP to remove and instead trims **real rallies** whose WASB trajectory registered only 0?1 net crossings:

```
| `min_crossings` | mean F1 | precision | recall | real-rally windows lost | ?F1 |
|---|---|---|---|---|---|
| 0 (off = `wasb_hybrid`) | 0.300 | 0.317 | 0.290 | 0 | ? |
| 1 | 0.303 | 0.327 | 0.288 | 1 | +0.003 |
| 2 (shipped #146) | 0.292 | 0.337 | 0.261 | 12 | ?0.008 |
| 3 | 0.269 | 0.338 | 0.226 | 28 | ?0.031 |
```

The shipped `mc=2` is a small mean-F1 **regression** (?0.008) and drops **12** windows that match golden rallies (worst on doubles, `largetest\_doubles` ?0.051). Honest caveats: deltas are **within n=6 noise** (not catastrophic), but `mc=2` is **not** a served improvement ? and because a dropped candidate **never reaches Gemini**, each lost rally is an **unrecoverable** served recall loss, while the precision "gain" partly duplicates what Gemini already does at the handoff. So the served tradeoff is **? neutral**.

Root cause: eval-harness windowing ? served windowing. Gate-A's value is conditional on over-segmentation; the coarse served windowing doesn't over-segment, so the gate inverts.

This is exactly the served-side follow-up the flip entry said was owed ? it doesn't unwind the merge, but the served numbers don't support `mc=2` as the default. The decision on what to do with the just-merged default is yours:

85. assistant (2026-07-02T06:11:33.158Z)

```
[external_agent_tool_call: AskUserQuestion]
input: {"questions":[{"question":"The Gate-A served default (min_crossings=2) shows a mild served-side regression on the production windowing. How do you want to handle the live default? (All are single-line config.json changes, no code.)","header":"Gate-A default","multiSelect":false,"options":[{"label":"Re-tune to mc=1","description":"Set indexing.fusion.min_crossings: 1 ? keeps Gate-A's intent (drops genuinely-dead 0-crossing feeds) at the best served operating point: ?flat F1 (+0.003), only 1 real rally lost, +0.010 precision. Preserves the held-shuttle guard without over-dropping real rallies."},{label":"Roll back to mc=0","description":"Set indexing.fusion.min_crossings: 0 ? Gate-A fully off; served path becomes byte-identical to wasb_hybrid (verified). Cleanest, zero served regression, but loses the held-shuttle guard entirely."},{label":"Leave mc=2 as-is","description":"Accept the ?0.008 (within n=6 noise); the offline 6/6 ablation + unit tests stand. Revisit when the golden corpus grows. I just keep the honest record in QUALITY_ITERATIONS.md."},{label":"Full revert segmenter","description":"Set indexing.default_segmenter: wasb_hybrid ? reverts the whole #146 served-segmenter flip, not just the gate threshold."}]}],"question":"Should I commit the QUALITY_ITERATIONS.md record (and any config change you pick above)?","header":"Commit","multiSelect":false,"options":[{"label":"Commit on a branch + PR","description":"Branch from origin/master, commit the doc record (+ config change if chosen), open one PR. I will NOT self-merge a quality-default change without your go-ahead."},{label":"Commit doc only, hold config","description":"Commit just the QUALITY_ITERATIONS.md honest record now; apply the config decision separately once you confirm."},{label":"Leave uncommitted","description":"Keep the doc edit in the working tree only; you'll handle the commit. Nothing pushed."}]}]}
[/external_agent_tool_call]
```

86. assistant (2026-07-02T06:11:33.158Z)

```
[external_agent_tool_result]
The user did not answer the questions.
[/external_agent_tool_result]
```

87. assistant (2026-07-02T06:11:33.158Z)

I'll proceed with the safe default: no unilateral change to a production quality-default, and no commit without your go-ahead. Here's where things stand.

Done

- **Served-side verification ran** (offline, reusing cached golden trajectories ? no WASB re-run, no GPU, no Gemini).
- **(a) Confirmed**: at `mc=0`, `fusion_hybrid` is byte-identical to `wasb_hybrid` (seeding + handoff) on all 6 golden videos ? the no-regression invariant holds on the live served code.
- **(b) Surfaced honestly**: the offline `+0.074` win does **not** transfer to the served (production) windowing; the shipped `mc=2` is a within-noise mean-F1 regression (`?0.008`) that drops 12 real-rally windows. `mc=1` is the better served operating point.
- **Recorded** in `[docs/QUALITY_ITERATIONS.md]` (top of live track) and in the memory checkpoint (durable regardless of commit).

Working-tree state (nothing pushed, nothing committed)

- `docs/QUALITY_ITERATIONS.md` ? **modified, uncommitted** (the honest record).
- `config.json` ? **untouched**, still `min_crossings: 2`. I did not flip the default; that's your call.
- Repro scripts (`scratch/served_gateA_{verify,sweep}.py`) + artifacts (`output/gateA_served_verify/`) ? in gitignored dirs.

Your call (whenever ? not blocking)

The single-flag options are in the question above; my recommendation is `mc=1` (keeps the held-shuttle guard's intent ? drops 0-crossing dead feeds ? without over-dropping real 1-crossing rallies). The deeper fix worth a separate item: **align the experiment-harness windowing with the served windowing**, so a harness promotion actually predicts served behavior ? that mismatch is the real lesson here, beyond this one threshold.

Tell me which option you want (or "commit the doc") and I'll apply it on a branch + PR.

88. user (2026-07-02T06:11:33.158Z)

definitely commit the doc!

89. assistant (2026-07-02T06:11:33.158Z)

On it. Following the shared-checkout branch protocol (branch from `origin/master` explicitly, stage only the doc, re-verify HEAD before pushing). Let me fetch and check the remote first.

90. assistant (2026-07-02T06:11:33.158Z)

```
[external_agent_tool_call: Bash]
description: Fetch and verify remote state before branching
command: git fetch origin 2>&1 | tail -3
echo "--- current branch ---"; git branch --show-current
echo "--- local HEAD ---"; git log --oneline -1
echo "--- origin/master HEAD ---"; git log --oneline -1 origin/master
echo "--- did origin/master move / touch the doc? ---"; git log --oneline e75ec35..origin/master
-- docs/QUALITY_ITERATIONS.md 2>&1 | head
echo "--- working tree ---"; git status --short
[/external_agent_tool_call]
```

91. assistant (2026-07-02T06:11:33.159Z)

```
[external_agent_tool_result]
--- current branch ---
master
--- local HEAD ---
e75ec35 feat(eval): LOS0/LOG0 ablation runner ? per-signal marginal ?F1 (Phase 0) (#147)
--- origin/master HEAD ---
e75ec35 feat(eval): LOS0/LOG0 ablation runner ? per-signal marginal ?F1 (Phase 0) (#147)
--- did origin/master move / touch the doc? ---
--- working tree ---
M docs/QUALITY_ITERATIONS.md
?? artifacts/
[/external_agent_tool_result]
```

92. assistant (2026-07-02T06:11:33.159Z)

Clean: `origin/master == local HEAD == e75ec35`, no sibling commits, no remote changes to the doc. Branching from `origin/master` and committing the doc only.

93. assistant (2026-07-02T06:11:33.159Z)

```
[external_agent_tool_call: Bash]
description: Create branch from origin/master and stage only the doc
command: git checkout -b docs/served-gate-a-verification origin/master 2>&1 | tail -3
echo "--- staged path (explicit) ---"
git add docs/QUALITY_ITERATIONS.md
git status --short
[/external_agent_tool_call]
```

94. assistant (2026-07-02T06:11:33.159Z)

```
[external_agent_tool_result]
M      docs/QUALITY_ITERATIONS.md
branch 'docs/served-gate-a-verification' set up to track 'origin/master'.
Switched to a new branch 'docs/served-gate-a-verification'
--- staged path (explicit) ---
M  docs/QUALITY_ITERATIONS.md
?? artifacts/
[/external_agent_tool_result]
```

95. assistant (2026-07-02T06:11:33.159Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\commitmsg_servedgatea.txt (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

96. assistant (2026-07-02T06:11:33.159Z)

On the branch with only the doc staged (`artifacts/` untracked, not staged). Committing via a message file to avoid shell-quoting issues.

97. assistant (2026-07-02T06:11:33.159Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\commitmsg_servedgatea.txt
[/external_agent_tool_call]
```

98. assistant (2026-07-02T06:11:33.159Z)

```
[external_agent_tool_call: Bash]
description: Commit the doc and verify only my commit/file
command: git commit -F scratch/commitmsg_servedgatea.txt 2>&1 | tail -5
echo "--- re-verify: HEAD is mine, only my commit ahead of origin/master ---"
git branch --show-current
git log --oneline origin/master..HEAD
echo "--- files in the commit (must be ONLY the doc) ---"
git show --stat --oneline HEAD | head -10
[/external_agent_tool_call]
```

99. assistant (2026-07-02T06:11:33.159Z)

```
[external_agent_tool_result]
[docs/served-gate-a-verification f6acde5] docs(quality): served-side Gate-A check ? flip's
offline win doesn't transfer to production windowing
 1 file changed, 47 insertions(+)
--- re-verify: HEAD is mine, only my commit ahead of origin/master ---
```

```
docs/served-gate-a-verification
f6acde5 docs(quality): served-side Gate-A check ? flip's offline win doesn't transfer to
production windowing
--- files in the commit (must be ONLY the doc) ---
f6acde5 docs(quality): served-side Gate-A check ? flip's offline win doesn't transfer to
production windowing
  docs/QUALITY_ITERATIONS.md | 47 ++++++
  1 file changed, 47 insertions(+)
[/external_agent_tool_result]
```

100. assistant (2026-07-02T06:11:33.159Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\badminton-highlight-
indexer\scratch\prbody_servedgatea.md (file state is current in your context ? no need to Read
it back)
[/external_agent_tool_result]
```

101. assistant (2026-07-02T06:11:33.159Z)

Commit is clean: `f6acde5`, one file (`docs/QUALITY\_ITERATIONS.md`, +47), only my commit ahead of `origin/master`. Now pushing and opening a PR (docs-only, one work item).

102. assistant (2026-07-02T06:11:33.159Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\prbody_servedgatea.md
[/external_agent_tool_call]
```

103. assistant (2026-07-02T06:11:33.159Z)

```
[external_agent_tool_call: Bash]
description: Push branch and open PR
command: git push -u origin docs/served-gate-a-verification 2>&1 | tail -5
echo "--- create PR ---"
gh pr create --base master --head docs/served-gate-a-verification \
  --title "docs(quality): served-side Gate-A check ? flip's offline win doesn't transfer to
production windowing" \
  --body-file scratch/prbody_servedgatea.md 2>&1 | tail -5
[/external_agent_tool_call]
```

104. assistant (2026-07-02T06:11:33.159Z)

```
[external_agent_tool_result]
remote:      https://github.com/avidullu/badminton-highlight-indexer/pull/new/docs/served-gate-
a-verification
remote:
branch 'docs/served-gate-a-verification' set up to track 'origin/docs/served-gate-a-
verification'.
To https://github.com/avidullu/badminton-highlight-indexer.git
 * [new branch]      docs/served-gate-a-verification -> docs/served-gate-a-verification
--- create PR ---
https://github.com/avidullu/badminton-highlight-indexer/pull/152
[/external_agent_tool_result]
```

105. assistant (2026-07-02T06:11:33.159Z)

PR opened: [#152](https://github.com/avidullu/badminton-highlight-indexer/pull/152). Let me check CI status (gating explicitly on the status rollout, per the merge-discipline gotcha).

106. assistant (2026-07-02T06:11:33.159Z)

```
[external_agent_tool_call: Bash]
description: Check PR #152 CI rollout status
```

```
command: sleep 20; gh pr view 152 --json mergeable,mergeStateStatus,statusCheckRollup --jq
'{mergeable,mergeStateStatus,checks:[.statusCheckRollup[]|{name,status,conclusion}]}' 2>&1 |
head -40
[/external_agent_tool_call]
```

107. assistant (2026-07-02T06:11:33.159Z)

```
[external_agent_tool_result]
{"checks":[{"conclusion":"SUCCESS","name":"Ruff
lint","status":"COMPLETED"}, {"conclusion":"","name":"Mypy (lenient
baseline)","status":"IN_PROGRESS"}, {"conclusion":"SUCCESS","name":"Syntax sweep + import smoke
(no GPU stack)","status":"COMPLETED"}, {"conclusion":"","name":"Full test suite (offline,
mocked)","status":"IN_PROGRESS"}], "mergeStateStatus":"UNKNOWN", "mergeable":"UNKNOWN"}
[/external_agent_tool_result]
```

108. assistant (2026-07-02T06:11:33.160Z)

CI running (2 of 4 green, mypy + full suite in progress). Let me wait for it to finish.

109. assistant (2026-07-02T06:11:33.160Z)

```
[external_agent_tool_call: Bash]
description: Re-check PR #152 CI after waiting
command: sleep 110; gh pr view 152 --json statusCheckRollup --jq
'[,.statusCheckRollup[]|{name,status,conclusion}]' 2>&1 | head -40
[/external_agent_tool_call]
```

110. assistant (2026-07-02T06:11:33.160Z)

```
[external_agent_tool_result: error]
<tool_use_error>Blocked: sleep 110 followed by: gh pr view 152 --json statusCheckRollup --jq
'[,.statusCheckRollup[]|{name,status,conclusion}]' head -40. To wait for a condition, use Monitor
with an until-loop (e.g. `until <check>; do sleep 2; done`). To wait for a command you started,
use run_in_background: true. Do not chain shorter sleeps to work around this
block.</tool_use_error>
[/external_agent_tool_result]
```

111. assistant (2026-07-02T06:11:33.160Z)

```
[external_agent_tool_call: Bash]
description: Re-query PR #152 status now
command: gh pr view 152 --json mergeable,mergeStateStatus,statusCheckRollup --jq
'{mergeable,mergeStateStatus,checks:[.statusCheckRollup[]|{name,conclusion}]}' 2>&1 | head -40
[/external_agent_tool_call]
```

112. assistant (2026-07-02T06:11:33.160Z)

```
[external_agent_tool_result]
{"checks":[{"conclusion":"SUCCESS","name":"Ruff lint"}, {"conclusion":"SUCCESS","name":"Mypy
(lenient baseline)"}, {"conclusion":"SUCCESS","name":"Syntax sweep + import smoke (no GPU
stack)"}, {"conclusion":"","name":"Full test suite (offline,
mocked)"}], "mergeStateStatus":"UNSTABLE", "mergeable":"MERGEABLE"}
[/external_agent_tool_result]
```

113. assistant (2026-07-02T06:11:33.160Z)

3 of 4 green; full test suite still running (`UNSTABLE` = pending check). I won't merge until all are SUCCESS. Let me wait on it with an until-loop.

114. assistant (2026-07-02T06:11:33.160Z)

```
[external_agent_tool_call: Bash]
description: Poll until full test suite concludes, then show rollup
```

```

command: i=0
until [ -n "$(gh pr view 152 --json statusCheckRollup --jq
'.statusCheckRollup[]|select(.name|startswith("Full test"))|.conclusion')" ] || [ $i -ge 60 ];
do sleep 5; i=$((i+1)); done
echo "waited ~$((i*5))s"
gh pr view 152 --json mergeable,mergeStateStatus,statusCheckRollup --jq
'{mergeable,mergeStateStatus,checks:[.statusCheckRollup[]|{name,conclusion}]}'
[/external_agent_tool_call]

```

115. assistant (2026-07-02T06:11:33.160Z)

```

[external_agent_tool_result]
waited ~60s
{"checks":[{"conclusion":"SUCCESS","name":"Ruff lint"},{"conclusion":"SUCCESS","name":"Mypy
(lenient baseline)"}, {"conclusion":"SUCCESS","name":"Syntax sweep + import smoke (no GPU
stack)"}, {"conclusion":"SUCCESS","name":"Full test suite (offline,
mocked)"}], "mergeStateStatus":"CLEAN", "mergeable":"MERGEABLE"}
[/external_agent_tool_result]

```

116. assistant (2026-07-02T06:11:33.160Z)

All 4 CI lanes green, `mergeStateStatus: CLEAN`. This is a docs-only honest record ? within my standing authority to merge on verified-green CI, and you clearly want it landed. Merging (squash) and cleaning up.

117. assistant (2026-07-02T06:11:33.160Z)

```

[external_agent_tool_call: Bash]
description: Merge PR, return to master, pull, verify, tidy temp files
command: gh pr merge 152 --squash --delete-branch 2>&1 | tail -5
echo "--- back to master, ff pull ---"
git checkout master 2>&1 | tail -2
git pull --ff-only 2>&1 | tail -3
echo "--- verify landed + clean ---"
git log --oneline -2
git status --short
echo "--- tidy temp commit/PR message files ---"
rm -f scratch/commitmsg_servedgatea.txt scratch/prbody_servedgatea.md && echo "removed temp msg
files (repro scripts kept)"
[/external_agent_tool_call]

```

118. assistant (2026-07-02T06:11:33.160Z)

```

[external_agent_tool_result]
tests/test_fusion_hybrid.py          | 86 +-----
tests/test_served_gate_a_regression.py | 160 +++++
8 files changed, 478 insertions(+), 157 deletions(-)
create mode 100644 backend/eval/served_gate_a.py
create mode 100644 tests/test_served_gate_a_regression.py
--- back to master, ff pull ---
Your branch is up to date with 'origin/master'.
Already on 'master'
Already up to date.
--- verify landed + clean ---
53bc743 docs(quality): served-side Gate-A check ? flip's offline win doesn't transfer to
production windowing (#152)
2cd469e revert(fusion): undo the Gate-A served-default flip (#146) ? confirmed served regression
(eval?serve) (#151)
?? artifacts/
--- tidy temp commit/PR message files ---
removed temp msg files (repro scripts kept)
[/external_agent_tool_result]

```

119. assistant (2026-07-02T06:11:33.160Z)

Important: while my docs PR was in flight, a ****sibling session merged** [#151](https://github.com/avidullu/badminton-highlight-indexer/pull/151)****** ? a revert of the Gate-A flip that independently reached the same conclusion ("confirmed served regression (eval?serve)") and even added a `backend/eval/served\_gate\_a.py` module + regression test. My docs PR #152 landed cleanly on top (`53bc743`, CI green). Let me reconcile ? my doc entry currently presents the rollback as still-pending, which may now be stale. Let me see what #151 actually did.

120. assistant (2026-07-02T06:11:33.160Z)

```
[external_agent_tool_call: Bash]
description: Inspect #151 revert effect on config and its diff
command: echo "=== config.json now (did #151 flip mc back?) ===" && python -c "import
json;f=json.load(open('config.json'))['indexing'];print('default_segmenter
=',f['default_segmenter']);print('fusion.min_crossings =',f['fusion']['min_crossings'])"
echo; echo "=== #151 commit body ===" && git log -1 2cd469e --format='%s%n%n%b' | head -40
echo; echo "=== #151 files changed ===" && git show --stat --oneline 2cd469e | head -20
[/external_agent_tool_call]
```

121. assistant (2026-07-02T06:11:33.160Z)

```
[external_agent_tool_result]
=== config.json now (did #151 flip mc back?) ===
Traceback (most recent call last):
  File "<string>", line 1, in <module>
    import json;f=json.load(open('config.json'))['indexing'];print('default_segmenter
=',f['default_segmenter']);print('fusion.min_crossings =',f['fusion']['min_crossings'])
~^^^^^^^^^^^^^^
```

```
KeyError: 'fusion'
default_segmenter = gemini
```

```
=== #151 commit body ===
revert(fusion): undo the Gate-A served-default flip (#146) ? confirmed served regression
(eval?serve) (#151)
```

Restore the pre-#146 no-regression production state. The #146 flip promoted Gate A (net-crossings ≥ 2) from a recommended config to the live served default, on the strength of an offline $+0.074$?F1. A served-side check (the #148 litmus + an independent served run) found the lift does NOT transfer: on the production COARSE windowing Gate A is neutral-to-negative (mean served ?F1 -0.008 , ~12 real rallies dropped, 2/6 losses).

Root cause (eval?serve, not a bug): the $+0.074$ is a property of the offline harness's permissive PROPOSAL windowing ($vt=0.005$, $gap=1.0$, $min=0.5$), which emits many short windows with held-shuttle junk for Gate A to cut. The served COARSE windowing ($vt=0.02$, $gap=2.0$, $min=2.0$) already under-segments (over-seg ratio < 1), so there is little junk left and the gate trims borderline true windows instead.

Reverted (back to the pre-#146 state):

- config.json: indexing.default_segmenter fusion_hybrid -> gemini (the prior served detector); the #146-added indexing.fusion block removed.
- backend/config/models.py: FusionConfig.min_crossings default 2 -> 0 (Gate A OFF by default). Gate A is NOT deleted ? it stays opt-in (indexing.fusion.min_crossings: 2).
- tests/test_config.py, tests/test_fusion_hybrid.py: restored to the Gate-A-OFF defaults (test_fusion_hybrid.py is byte-identical to its pre-#146 form). The persisted-superset invariant is preserved ? gating only ever filtered the AI handoff, never persistence.
- docs/QUALITY_ITERATIONS.md: the Gate-A SERVED-DEFAULT entry now reads REVERTED with the honest eval?serve reason; the PROMOTED entry's stale "owner approved the flip" pointer is corrected.
- eval_baselines/experiment_leaderboard.json: the #146 promotion record carries a "reverted (eval?serve)" note.

The #148 served litmus (backend/eval/served_gate_a.py + tests/test_served_gate_a_regression.py) is the standing guard that caught this and pins the honest measured behaviour.

ruff + mypy clean; full suite 605 passed, 2 skipped (golden-data-gated).

```
=== #151 files changed ===
2cd469e revert(fusion): undo the Gate-A served-default flip (#146) ? confirmed served regression
(eval?serve) (#151)
  backend/config/models.py          | 33 +++-----
  config.json                      |  8 +-
  docs/QUALITY_ITERATIONS.md       | 63 ++++++++-----
  eval_baselines/experiment_leaderboard.json |  1 +
  tests/test_config.py             | 38 +++++-----
  tests/test_fusion_hybrid.py      | 86 +-----
  6 files changed, 72 insertions(+), 157 deletions(-)
[/external_agent_tool_result]
```

122. assistant (2026-07-02T06:11:33.160Z)

Now I have the full picture. A sibling session did the complete rollback in parallel:

- **PR #148** productionized a served-side litmus: `backend/eval/served\_gate\_a.py` + `tests/test\_served\_gate\_a\_regression.py` (standing guard).
- **PR #151** reverted #146: `config.json` ? `default\_segmenter: gemini` (the `indexing.fusion` block removed ? that's why my probe hit `KeyError: 'fusion'`), `FusionConfig.min\_crossings` default 2?0 (Gate A now opt-in), and it already updated the SERVED-DEFAULT doc entry to "REVERTED". Its independent numbers **match mine** (20.008, ~12 rallies dropped, 2/6 losses).

So my just-merged entry is now **stale** ? it says "config.json untouched (mc=2 stays)" and frames the rollback as pending, when it's already done. That misstates the live state. I need to reconcile the doc to keep it honest. Let me read the current top sections to fix it precisely without contradicting #151's wording.

123. assistant (2026-07-02T06:11:33.160Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\QUALITY_ITERATIONS.md
[/external_agent_tool_call]
```

124. assistant (2026-07-02T06:11:33.160Z)

```
[external_agent_tool_result]
43     ### Served-side Gate-A check on the LIVE FusionSegmenter ? the flip's offline win does
NOT transfer to the production windowing *(2026-06-14)*
44     The owner-gated follow-up flagged by the SERVED-DEFAULT entry below: confirm the live
`FusionSegmenter` served path has no
45     regression from the Gate-A default flip (PR #146). Driven the real served seams
(`_enrich_candidate_stats` ?
46     `_net_crossings`, `_select_for_handoff` ? the gate) over each golden video's cached
WASB trajectory, at the production
47     windowing process_video actually uses (`velocity_thresh=0.02, merge_gap=2.0,
min_window=2.0` from config) ? no WASB
48     re-run (the runner has no cache-hit early-return), no GPU, no Gemini. Config loaded
exactly as production (`load_config()`);
49     segmenters from the registry. Repro: `scratch/served_gateA_verify.py` +
`scratch/served_gateA_sweep.py`; artifacts isolated
50     in `output/gateA_served_verify/`.
51
52     - (a) No-regression invariant CONFIRMED on real served code. At `min_crossings = 0`,
`fusion_hybrid`'s seeding and
53     handoff set are byte-identical to `wasb_hybrid` on all 6 golden videos (same
candidate windows; handoff = all windows =
54     the base seam). The "fusion adds nothing when the gates are off" property is now
verified on the live served path, not
55     only by code review. ?
56     - (b) The +0.074 win does NOT transfer to the served windowing. The offline
promotion (+0.074, 6/6, p=0.031) was measured
57     on the eval harness's recall-oriented proposal windowing `(0.005, 1.0, 0.5)`,
```

where CONTROL ****over-segments****

58 (segment-count ratio 1.68) and Gate-A's job is to clean up that fragmentation (?
1.01). The ****served path uses a coarse**
59 **windowing**** `(0.02, 2.0, 2.0)` that already merges aggressively (low over-
segmentation), so Gate-A has little FP to remove
60 and instead trims ****real**** rallies whose WASB trajectory registered only 0?1 net
crossings. ****Candidate-handoff-level F1**
61 (pre-Gemini, IoU ? 0.5) on the served windowing, n=6:
62
63 | min_crossings | mean F1 | precision | recall | real-rally windows lost vs off | ?F1
vs off |
64 |---|---|---|---|---|---|
65 | 0 (Gate-A OFF = `wasb\_hybrid`) | 0.300 | 0.317 | 0.290 | 0 | ? |
66 | 1 | ****0.303**** | 0.327 | 0.288 | 1 | ****+0.003**** |
67 | ****2 (shipped default, PR #146)**** | 0.292 | 0.337 | 0.261 | ****12**** | ****?0.008**** |
68 | 3 | 0.269 | 0.338 | 0.226 | 28 | ?0.031 |
69
70 At the shipped `mc=2`, mean F1 goes ****0.300 ? 0.292 (?0.008)**** and Gate-A drops ****12**
windows that match golden rallies
71 (net_crossings almost all exactly 1.0), concentrated in doubles (`largetest\_doubles`
?0.051). All deltas are ****tiny / within**
72 n=6 noise**, so this is not a catastrophic regression ? but `mc=2` is ****not**** a
served-side ****improvement****, and the
73 held-shuttle "guard" is removing genuine rallies, not just feeds.
74 - ****Served relevance (honest).**** A dropped candidate window ****never reaches Gemini****, so
each lost real rally is an
75 ****unrecoverable served recall loss****; the precision "gain", by contrast, is partly
what Gemini already does at the handoff ?
76 so the served-side tradeoff is at best neutral and plausibly net-negative, i.e. the
candidate-level ?0.008 likely
77 ****understates* the served downside.**
78 - ****Root cause = eval ? serve windowing.**** The harness promotes/refuses variants on a
windowing the production path does not
79 serve. Gate-A's value is conditional on over-segmentation; the coarse served windowing
doesn't over-segment, so the gate
80 inverts. [[decision-rigor-norm]]: the foundational issue (harness windowing ? served
windowing) outweighs the single
81 proposal-windowing number.
82
83 ****Recommendation (owner-gated, single-flag).**** `mc=1` is the better served operating
point if Gate-A stays on (?flat F1, only
84 1 real rally lost, +0.010 precision); `mc=0` is the clean rollback. Either is a one-line
config change (no code):
85 `indexing.fusion.min\_crossings: 1` or `: 0` ...
[/external_agent_tool_result]

125. assistant (2026-07-02T06:11:33.161Z)

Now I understand the full state: the doc now has my entry (stale ? says "mc=2 stays / decision pending") AND #151's authoritative "REVERTED" entry below it. My entry misstates the live state. I'll reconcile it to reflect reality (flip already reverted by #151, corroborated by my independent run) ? keeping its unique value (the byte-identity (a) confirmation + the mc-sweep) but fixing the stale framing.

126. assistant (2026-07-02T06:11:33.161Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\QUALITY_ITERATIONS.md
[/external_agent_tool_call]

127. assistant (2026-07-02T06:11:33.161Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\QUALITY_ITERATIONS.md has been

updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

128. assistant (2026-07-02T06:11:33.161Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\QUALITY_ITERATIONS.md
[/external_agent_tool_call]

129. assistant (2026-07-02T06:11:33.161Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\QUALITY_ITERATIONS.md has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

130. assistant (2026-07-02T06:11:33.161Z)

Now let me fix the memory checkpoint ? it also has the stale "decision pending / not committed yet" framing.

131. assistant (2026-07-02T06:11:33.161Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md
[/external_agent_tool_call]

132. assistant (2026-07-02T06:11:33.161Z)

[external_agent_tool_result]
The file C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

133. assistant (2026-07-02T06:11:33.161Z)

Memory updated. Now commit the doc reconciliation ? fetching first since sibling sessions are clearly active.

134. assistant (2026-07-02T06:11:33.161Z)

[external_agent_tool_call: Bash]
description: Fetch, branch from origin/master, stage doc
command: git fetch origin 2>&1 | tail -2
echo "--- local vs origin/master ---"; git log --oneline -1; git log --oneline -1 origin/master
echo "--- branch from origin/master ---"
git checkout -b docs/served-gate-a-reconcile origin/master 2>&1 | tail -2
git add docs/QUALITY_ITERATIONS.md
git status --short
[/external_agent_tool_call]

135. assistant (2026-07-02T06:11:33.161Z)

[external_agent_tool_result]
--- local vs origin/master ---
53bc743 docs(quality): served-side Gate-A check ? flip's offline win doesn't transfer to production windowing (#152)
53bc743 docs(quality): served-side Gate-A check ? flip's offline win doesn't transfer to production windowing (#152)
--- branch from origin/master ---
branch 'docs/served-gate-a-reconcile' set up to track 'origin/master'.

```
Switched to a new branch 'docs/served-gate-a-reconcile'
M docs/QUALITY_ITERATIONS.md
?? artifacts/
[/external_agent_tool_result]
```

136. assistant (2026-07-02T06:11:33.161Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\commitmsg_reconcile.txt
[/external_agent_tool_call]
```

137. assistant (2026-07-02T06:11:33.161Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\commitmsg_reconcile.txt (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

138. assistant (2026-07-02T06:11:33.161Z)

```
[external_agent_tool_call: Bash]
description: Commit, verify, push, open reconcile PR
command: git commit -F scratch/commitmsg_reconcile.txt 2>&1 | tail -3
echo "--- verify only my commit + only the doc ---"
git log --oneline origin/master..HEAD
git show --stat --oneline HEAD | head -6
echo "--- push + PR ---"
git push -u origin docs/served-gate-a-reconcile 2>&1 | tail -2
gh pr create --base master --head docs/served-gate-a-reconcile \
  --title "docs(quality): reconcile served-side Gate-A entry with the #151 revert" \
  --body "Follow-up to #152. My #152 entry was authored while the rollback looked pending; in parallel a sibling session had already productionized the served litmus (#148, \backend/eval/served_gate_a.py\) and **reverted** the #146 flip (#151). This corrects the now-stale framing so the living log stays honest: the entry is reframed as the **independent measurement corroborating** #148/#151 (numbers match: served ?F1 ?0.008, ~12 rallies dropped, 2/6 losses), and the 'decision pending' recommendation becomes 'Outcome ? REVERTED in #151' with the real config state + a cross-link. Measurement/tables and the (a) mc=0 byte-identity-to-\`wasb_hybrid\` confirmation are unchanged. Docs-only.

? Generated with [Claude Code](https://claude.com/claude-code)" 2>&1 | tail -3
rm -f scratch/commitmsg_reconcile.txt
[/external_agent_tool_call]
```

139. assistant (2026-07-02T06:11:33.161Z)

```
[external_agent_tool_result]
[docs/served-gate-a-reconcile 869c870] docs(quality): reconcile served-side Gate-A entry with the #151 revert
1 file changed, 19 insertions(+), 14 deletions(-)
--- verify only my commit + only the doc ---
869c870 docs(quality): reconcile served-side Gate-A entry with the #151 revert
869c870 docs(quality): reconcile served-side Gate-A entry with the #151 revert
docs/QUALITY_ITERATIONS.md | 33 ++++++-----
1 file changed, 19 insertions(+), 14 deletions(-)
--- push + PR ---
To https://github.com/avidullu/badminton-highlight-indexer.git
* [new branch] docs/served-gate-a-reconcile -> docs/served-gate-a-reconcile
https://github.com/avidullu/badminton-highlight-indexer/pull/153
[/external_agent_tool_result]
```

140. assistant (2026-07-02T06:11:33.161Z)

PR [#153](https://github.com/avidullu/badminton-highlight-indexer/pull/153) opened (docs-only, 1

file). Let me wait for CI and merge on green.

141. assistant (2026-07-02T06:11:33.161Z)

```
[external_agent_tool_call: Bash]
description: Poll until all CI checks conclude on PR #153
command: i=0
until [ "$(gh pr view 153 --json statusCheckRollup --jq
'[,.statusCheckRollup[]|select(.conclusion=="")]|length')" = "0" ] || [ $i -ge 72 ]; do sleep 5;
i=$((i+1)); done
echo "waited ~$((i*5))s"
gh pr view 153 --json mergeable,mergeStateStatus,statusCheckRollup --jq
'{mergeable,mergeStateStatus,checks:[.statusCheckRollup[]|{name,conclusion}]}'
[/external_agent_tool_call]
```

142. assistant (2026-07-02T06:11:33.161Z)

```
[external_agent_tool_result]
Command running in background with ID: bc7j0nc84. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\e03c8ada-c18d-40e6-9ee1-144846fe3975\tasks\bc7j0nc84.output. You will be notified when
it completes. To check interim output, use Read on that file path.
[/external_agent_tool_result]
```

143. assistant (2026-07-02T06:11:33.161Z)

CI for the reconcile PR is running in the background; I'll merge it once it goes green. Quick status while that finishes:

****What happened:**** Your "commit the doc" landed as [PR #152](https://github.com/avidullu/badminton-highlight-indexer/pull/152) (merged, `53bc743`, CI green). But I then discovered a ****sibling session** had already acted on the same finding in parallel******:

- ****[#148](https://github.com/avidullu/badminton-highlight-indexer/pull/148)**** productionized a served-side litmus (`backend/eval/served\_gate\_a.py` + a regression test).
- ****[#151](https://github.com/avidullu/badminton-highlight-indexer/pull/151)**** ****reverted the #146 flip**** ? `config.json` `default\_segmenter` ? `gemini`, the `indexing.fusion` block removed, `FusionConfig.min\_crossings` default `2?0` (Gate A now OFF-by-default but still opt-in).

Their independent numbers ****match mine**** (served ?F1 ?0.008, ~12 rallies dropped, 2/6 losses) ? good corroboration that the eval?serve regression is real.

That made my freshly-merged #152 entry ****stale**** (it framed the rollback as "decision pending / mc=2 stays"). So [PR #153](https://github.com/avidullu/badminton-highlight-indexer/pull/153) reconciles it: reframed as the independent measurement corroborating #148/#151, "Outcome ? REVERTED" instead of "decision pending", pointing at the productionized guard. Memory checkpoint updated to match.

****Net live state:**** the Gate-A served-default flip is ****already reverted**** ? your original question (rollback to mc=0 / retune to mc=1 / leave mc=2) is effectively moot; production is back on the pre-#146 `gemini` default with Gate A opt-in. The one genuinely open item is the deeper lesson: ****align the experiment-harness windowing with the served windowing**** so a harness promotion actually predicts served behavior.

I'll confirm #153 merged once CI completes.

144. user (2026-07-02T06:11:33.161Z)

```
<task-notification>
<task-id>bc7j0nc84</task-id>
<tool-use-id>toolu_01XFNrchpiwuHWCfpJ1cj7q</tool-use-id>
<output-file>C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-
highlight-indexer\e03c8ada-c18d-40e6-9ee1-144846fe3975\tasks\bc7j0nc84.output</output-file>
<status>completed</status>
```

```
<summary>Background command "Poll until all CI checks conclude on PR #153" completed (exit code 0)</summary>
</task-notification>
```

145. assistant (2026-07-02T06:11:33.161Z)

```
[external_agent_tool_call: Bash]
description: Confirm PR #153 CI rollup
command: gh pr view 153 --json mergeable,mergeStateStatus,statusCheckRollup --jq
'{mergeable,mergeStateStatus,checks:[.statusCheckRollup[]|{name,conclusion}]}'
[/external_agent_tool_call]
```

146. assistant (2026-07-02T06:11:33.162Z)

```
[external_agent_tool_result]
{"checks":[{"conclusion":"SUCCESS","name":"Ruff lint"}, {"conclusion":"SUCCESS","name":"Mypy (lenient baseline)"}, {"conclusion":"SUCCESS","name":"Syntax sweep + import smoke (no GPU stack)"}, {"conclusion":"SUCCESS","name":"Full test suite (offline, mocked)"}], "mergeStateStatus":"CLEAN", "mergeable":"MERGEABLE"}
[/external_agent_tool_result]
```

147. assistant (2026-07-02T06:11:33.162Z)

All 4 lanes green, `CLEAN`. Merging the reconcile PR and returning to a clean master.

148. assistant (2026-07-02T06:11:33.162Z)

```
[external_agent_tool_call: Bash]
description: Merge #153, return to master, verify final state
command: gh pr merge 153 --squash --delete-branch 2>&1 | tail -3
git checkout master 2>&1 | tail -1
git pull --ff-only 2>&1 | tail -2
echo "--- final state ---"
git log --oneline -3
git status --short
echo "--- live config sanity (post-#151 revert) ---"
python -c "import json;f=json.load(open('config.json'))['indexing'];print('default_segmenter =',f['default_segmenter'],'| fusion block present:', 'fusion' in f)"
[/external_agent_tool_call]
```

149. assistant (2026-07-02T06:11:33.162Z)

```
[external_agent_tool_result]
Fast-forward
 docs/QUALITY_ITERATIONS.md | 33 ++++++-----
 1 file changed, 19 insertions(+), 14 deletions(-)
Already on 'master'
Already up to date.
--- final state ---
a03f0f4 docs(quality): reconcile served-side Gate-A entry with the #151 revert (#153)
53bc743 docs(quality): served-side Gate-A check ? flip's offline win doesn't transfer to production windowing (#152)
2cd469e revert(fusion): undo the Gate-A served-default flip (#146) ? confirmed served regression (eval?serve) (#151)
?? artifacts/
--- live config sanity (post-#151 revert) ---
default_segmenter = gemini | fusion block present: False
[/external_agent_tool_result]
```

150. assistant (2026-07-02T06:11:33.162Z)

Done ? both PRs merged, master clean at `a03f0f4`, live config confirmed back to the pre-#146 state (`default\_segmenter: gemini`, no `fusion` block, Gate A off-by-default).

Final outcome

****Served-side verification (the task):****

- **** (a) No-regression invariant ? CONFIRMED ?**** At ``min_crossings=0``, the live ``FusionSegmenter`` seeding + handoff are byte-identical to ``wasb_hybrid`` on all 6 golden videos ? verified on real served code, not just by review.
- **** (b) Regression found ??**** The offline ``+0.074`` win does ****not**** transfer to the production (coarse) windowing. At the shipped ``mc=2``: mean F1 `?0.008`, ****12 real-rally windows dropped**** (worst on doubles). Root cause: ****eval-harness windowing ? served windowing**** ? Gate-A's value was an artifact of the over-segmenting `*proposal*` windowing the harness uses but production doesn't serve.

****What landed:****

- **[#152](https://github.com/avidullu/badminton-highlight-indexer/pull/152)** ? the honest measurement recorded in ``docs/QUALITY_ITERATIONS.md`` (your "commit the doc").
- **[#153](https://github.com/avidullu/badminton-highlight-indexer/pull/153)** ? reconciled that entry with reality, since a sibling session acted on the same finding in parallel.

****The parallel-session twist (worth knowing):**** while I was verifying, other slates independently reached the same conclusion and went further ?

****[#148](https://github.com/avidullu/badminton-highlight-indexer/pull/148)**** productionized a served-side litmus (``backend/eval/served_gate_a.py`` + regression test) and ****[#151](https://github.com/avidullu/badminton-highlight-indexer/pull/151)**** reverted the flip. Their numbers match mine (`?0.008`, ~12 dropped, 2/6 losses) ? independent corroboration. So the rollback decision I queued for you is ****already executed****; nothing for you to decide there.

****One genuinely open item (the real lesson, not requested ? flagging only):**** the experiment harness promotes/refuses cues on a windowing production doesn't serve, so a harness lift doesn't reliably predict served behavior. Aligning the harness windowing with the served windowing (or re-running promotions on production windowing) would prevent the next eval?serve surprise. I can scope that as its own work item whenever you want ? it's the kind of thing worth a proper design pass.